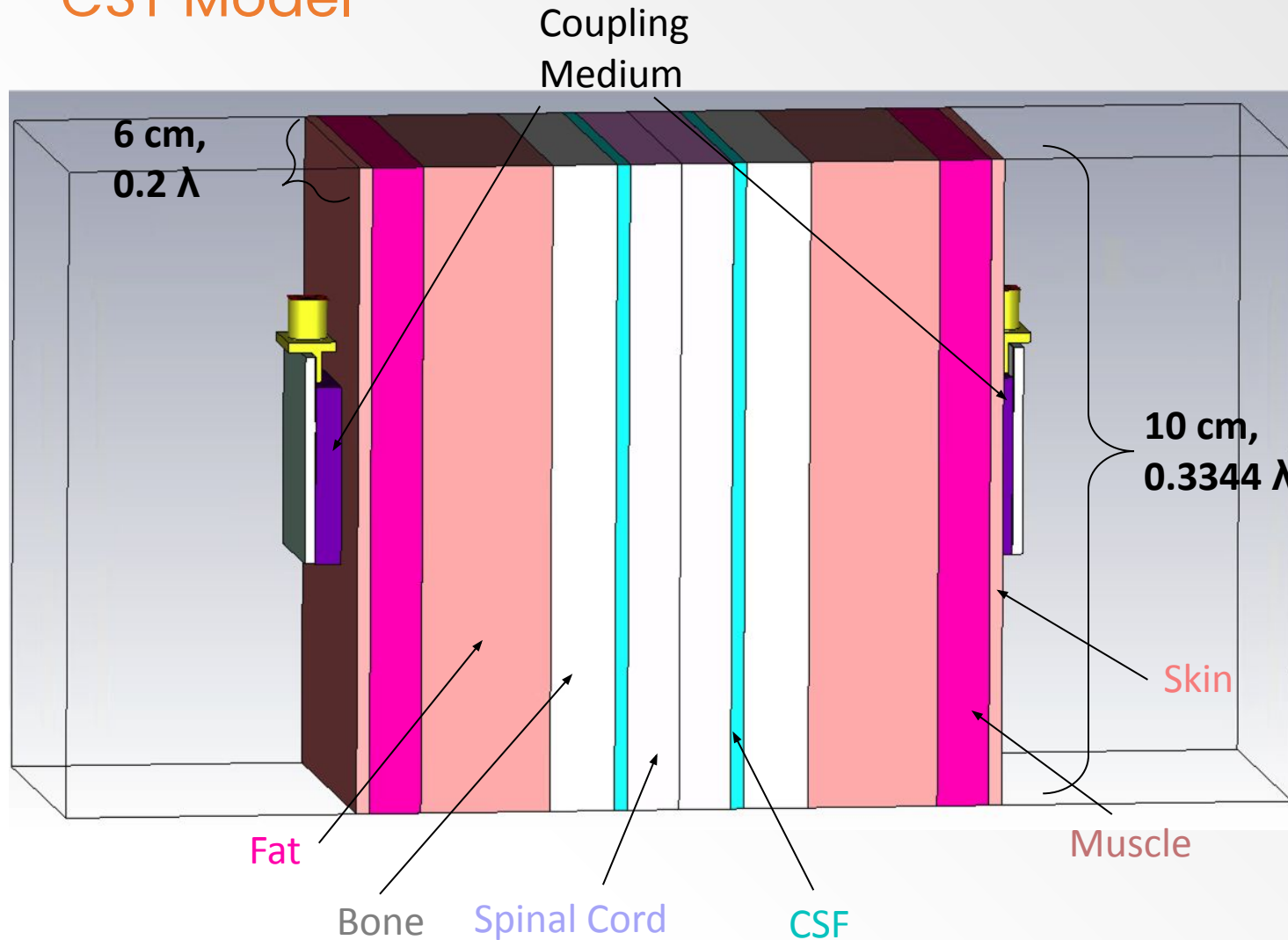


Design and Manufacturing of Low-Cost Antennas for Early Alzheimer's Detection via Medical Microwave Sensing of the Neck

Alex Peirano

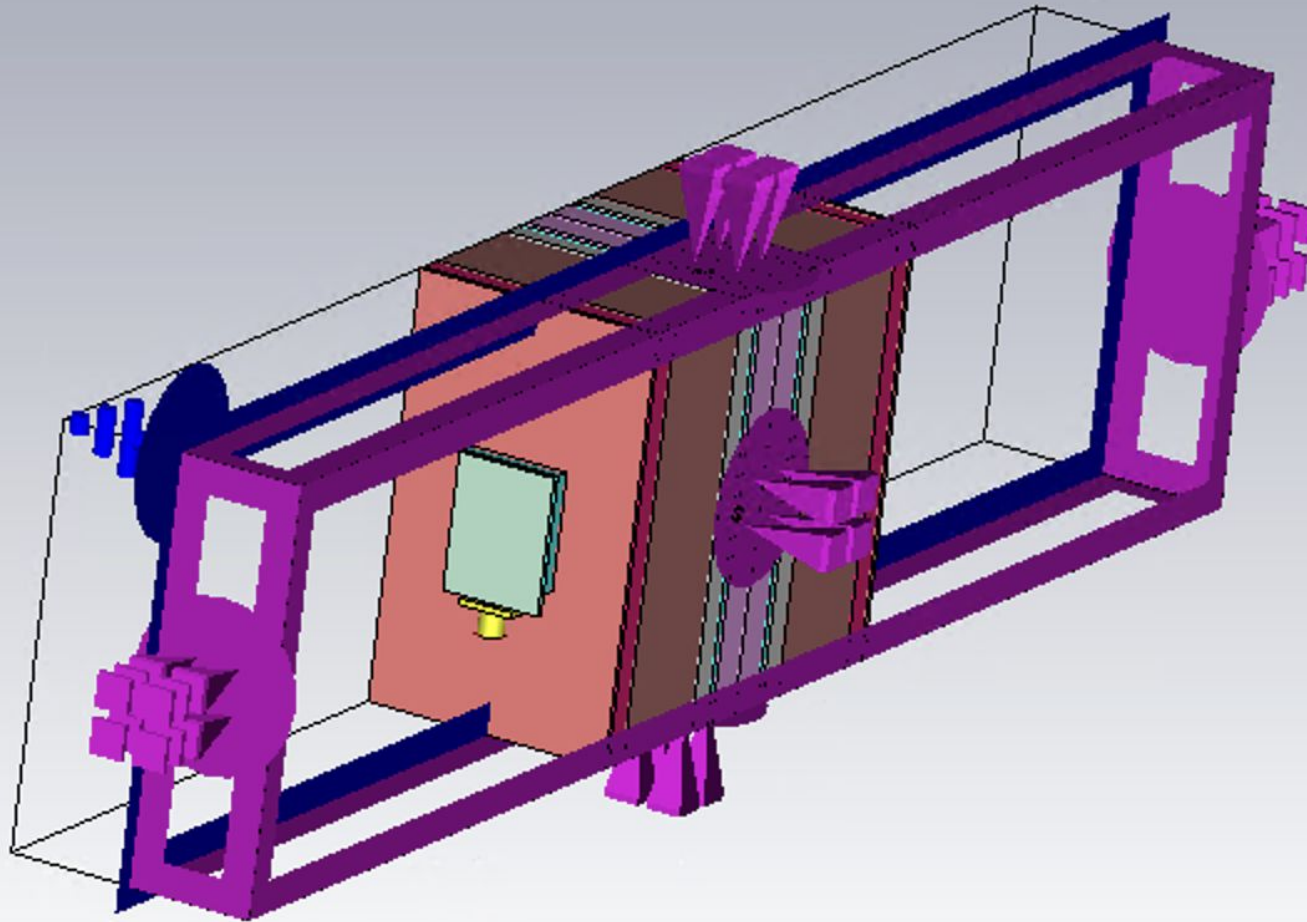
MultiTissue block

CST Model



LAYERS	THICKNESS	/ λ
Coupling Medium	0.4 cm	0.013
Skin	0.14 cm	0.005
Fat	0.67 cm	0.022
Muscle	3.0 cm	0.1
Bone	0.6925 cm	0.023
CSF	0.22 cm	0.007
Spinal Cord	1 cm	0.033

Simulations Boundary conditions



Boundary Conditions

Boundaries | Symmetry Planes

☐ Apply in all directions

Xmin: open Xmax: open

Ymin: open Ymax: open

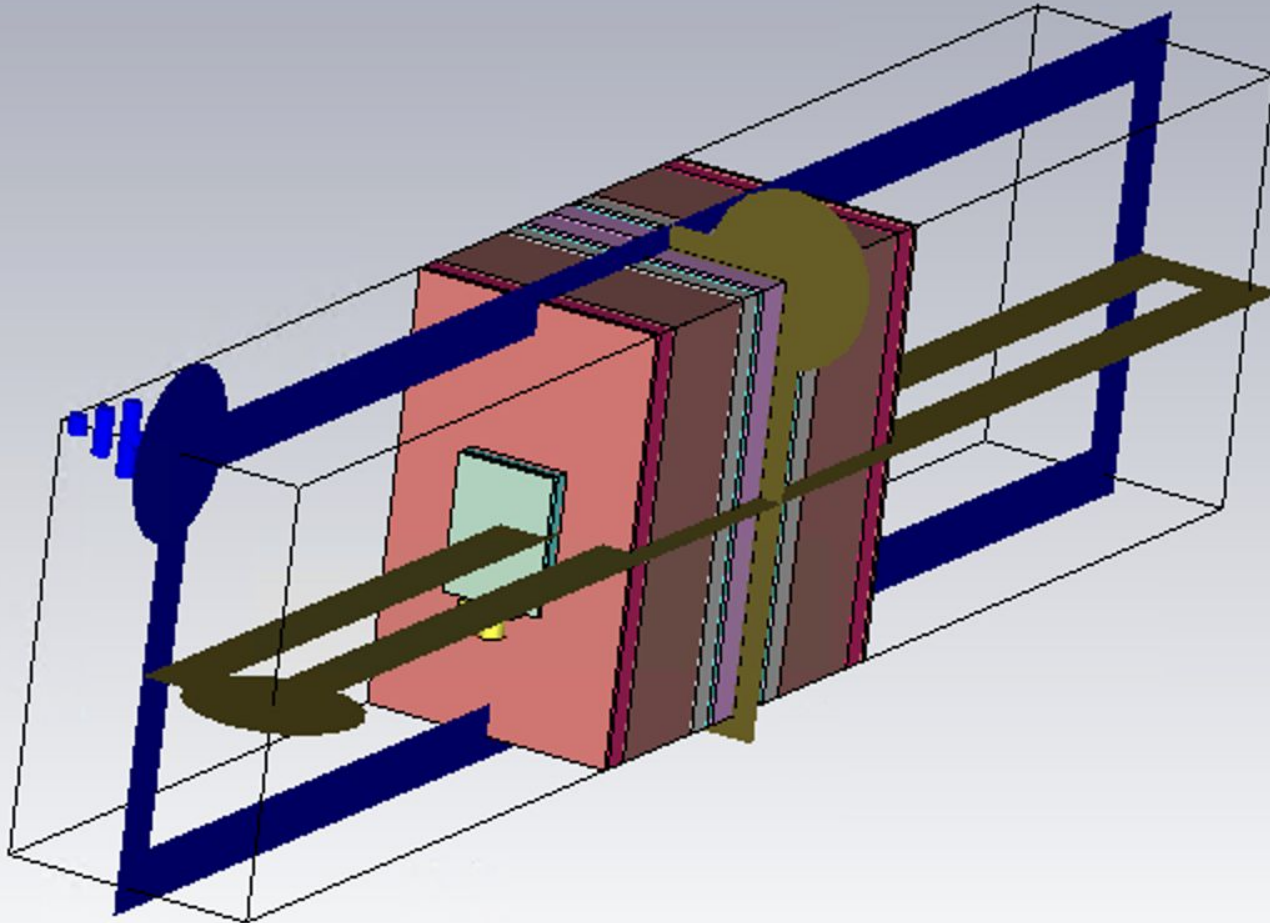
Zmin: open (add space) Zmax: open (add space)

Cond.: 1000 S/m Open Boundary...

OK Cancel Help



Simulations Boundary conditions



Boundary Conditions

Boundaries Symmetry Planes

YZ plane: magnetic ($H_t = 0$)

XZ plane: none

XY plane: none

OK Cancel Help



Simulations

Refining the mesh

Mesh Properties - Hexahedral FIT

Maximum cell

Cells per wavelength: Near to model: 10 Far from model: 10

☒ Use same setting as near to model

Cells per max model box edge: 20 1

☐ Use same setting as near to model

Minimum cell

Fraction of maximum cell near to model: 10

☒ Use same setting in all three directions

Statistics

Smallest cell:	Nx:
0.605	34
Largest cell:	Ny:
8.54449	103
Number of cells:	Nz:
356,796	107

OK

Cancel

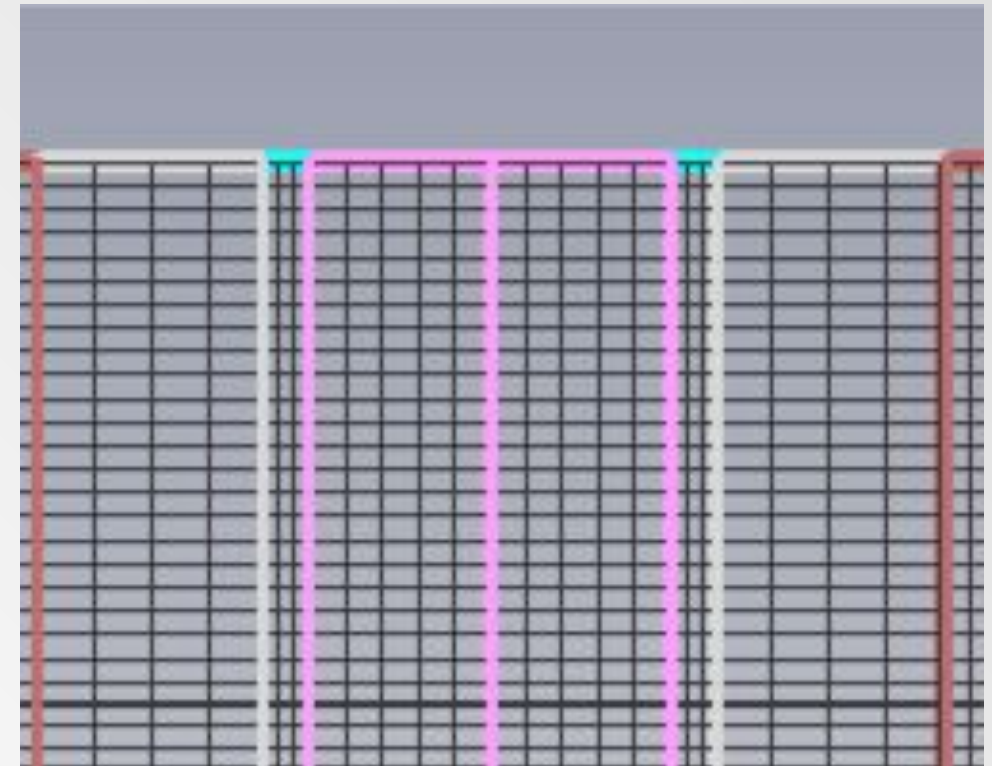
Apply

Update

Specials...

Simplify Model...

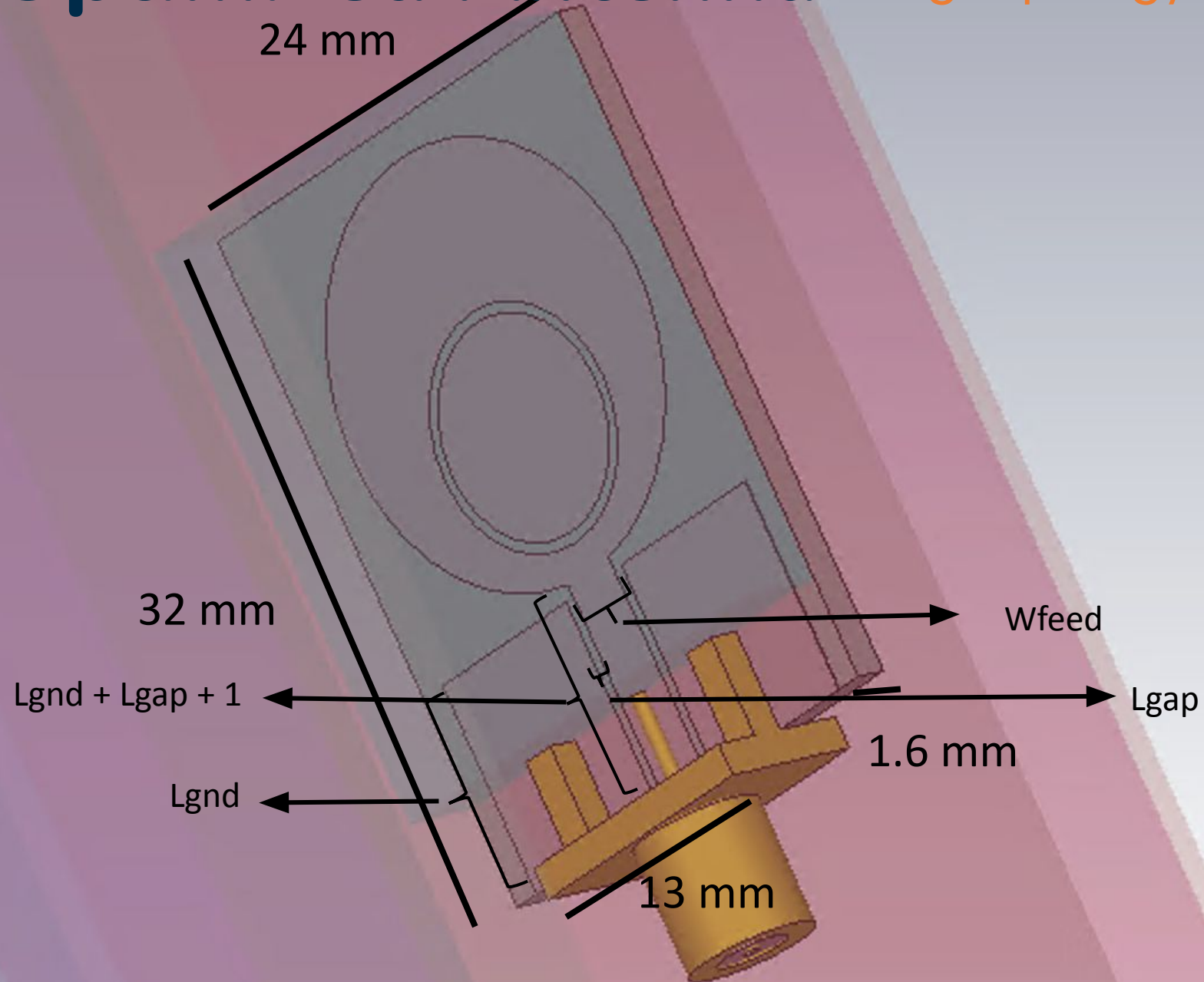
Help



3x increase in
the number
of cells

Optimized Antenna

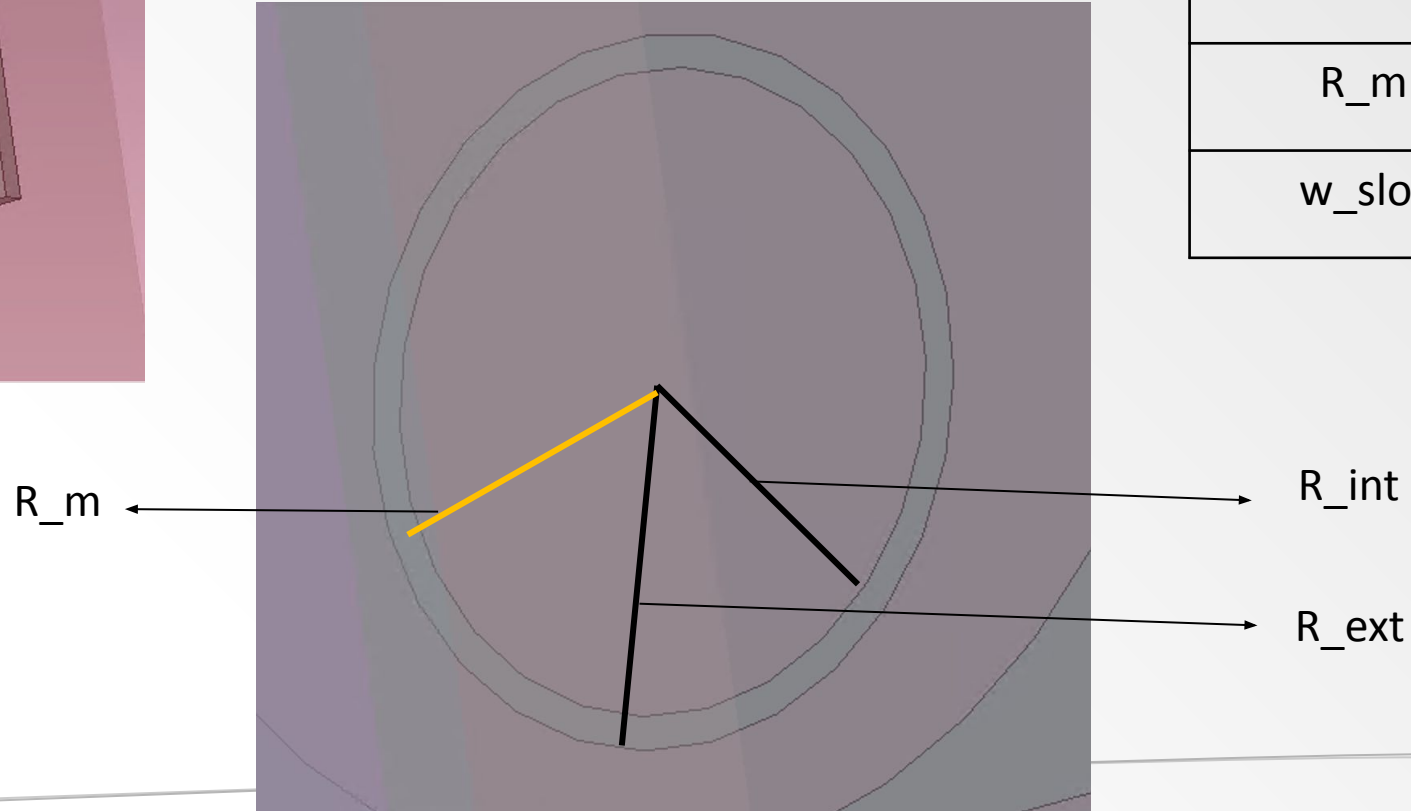
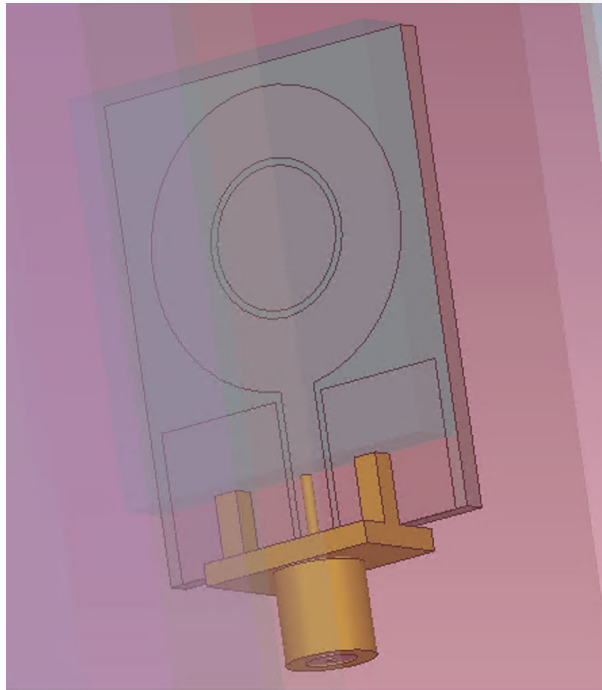
Ring topology



Thickness of the copper:
0.035mm

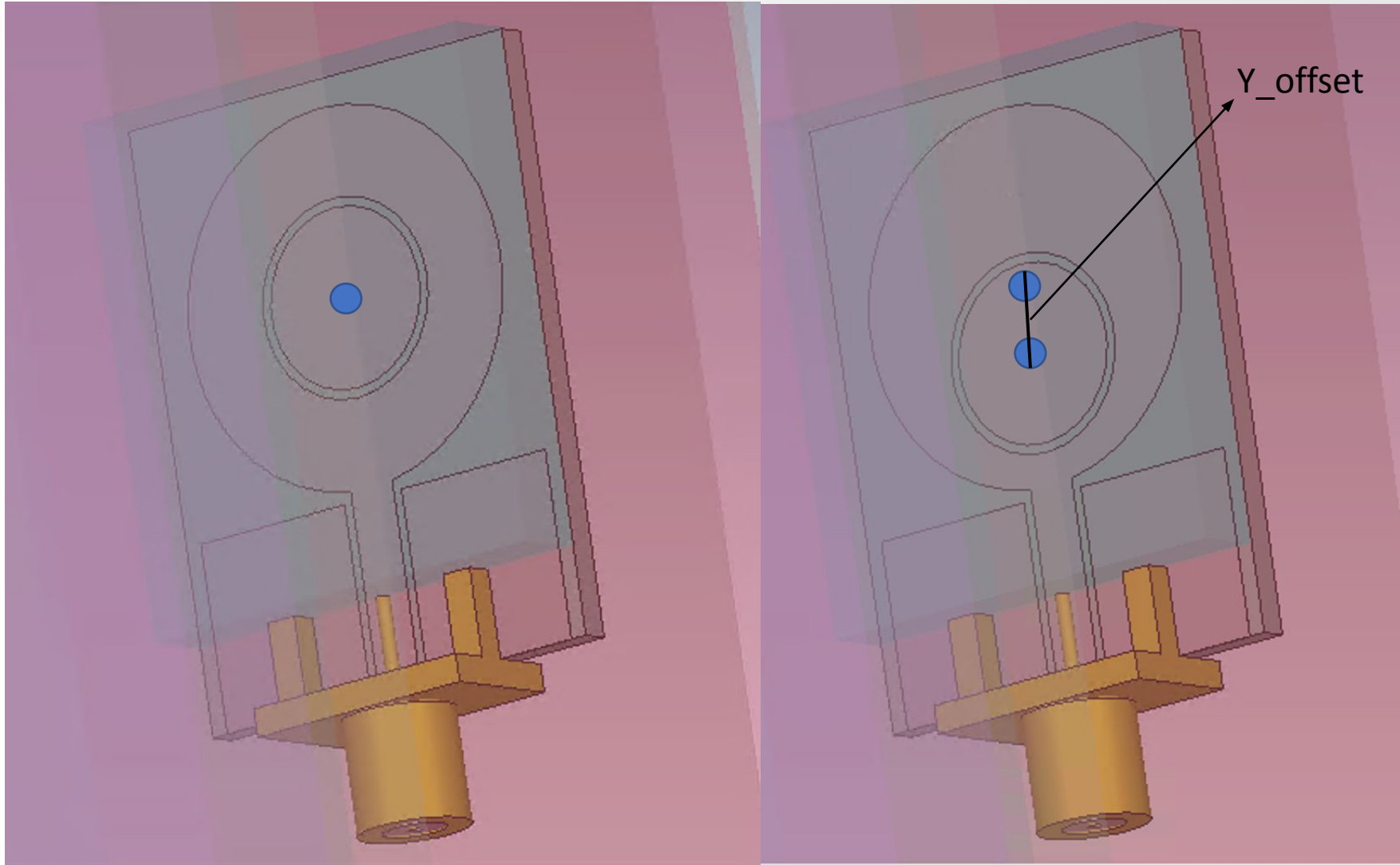
VARIABLES	DIMENSION
wfeed	2.9 mm
Lgap	0.5 mm
Wgnd	10 mm
Lgnd	10 mm

Optimized Antenna Ring topology



LAYERS	DIMENSION
R_{ext}	$R_m - w_{slot}/2$
R_{int}	$R_m + w_{slot}/2$
R_m	4.03 mm
w_{slot}	0.5 mm

Optimized Antenna Ring topology



LAYERS	DIMENSION
R_ext	5.25 mm
R_int	4.75 mm
R_m	5 mm
w_slot	0.5 mm
y_offset	1 mm

Preliminary computations

Ring Topology

Annular Ring Resonance Condition

$$2\pi r = n\lambda_g$$

For the fundamental mode ($n = 0.5$ for a half-wavelength):

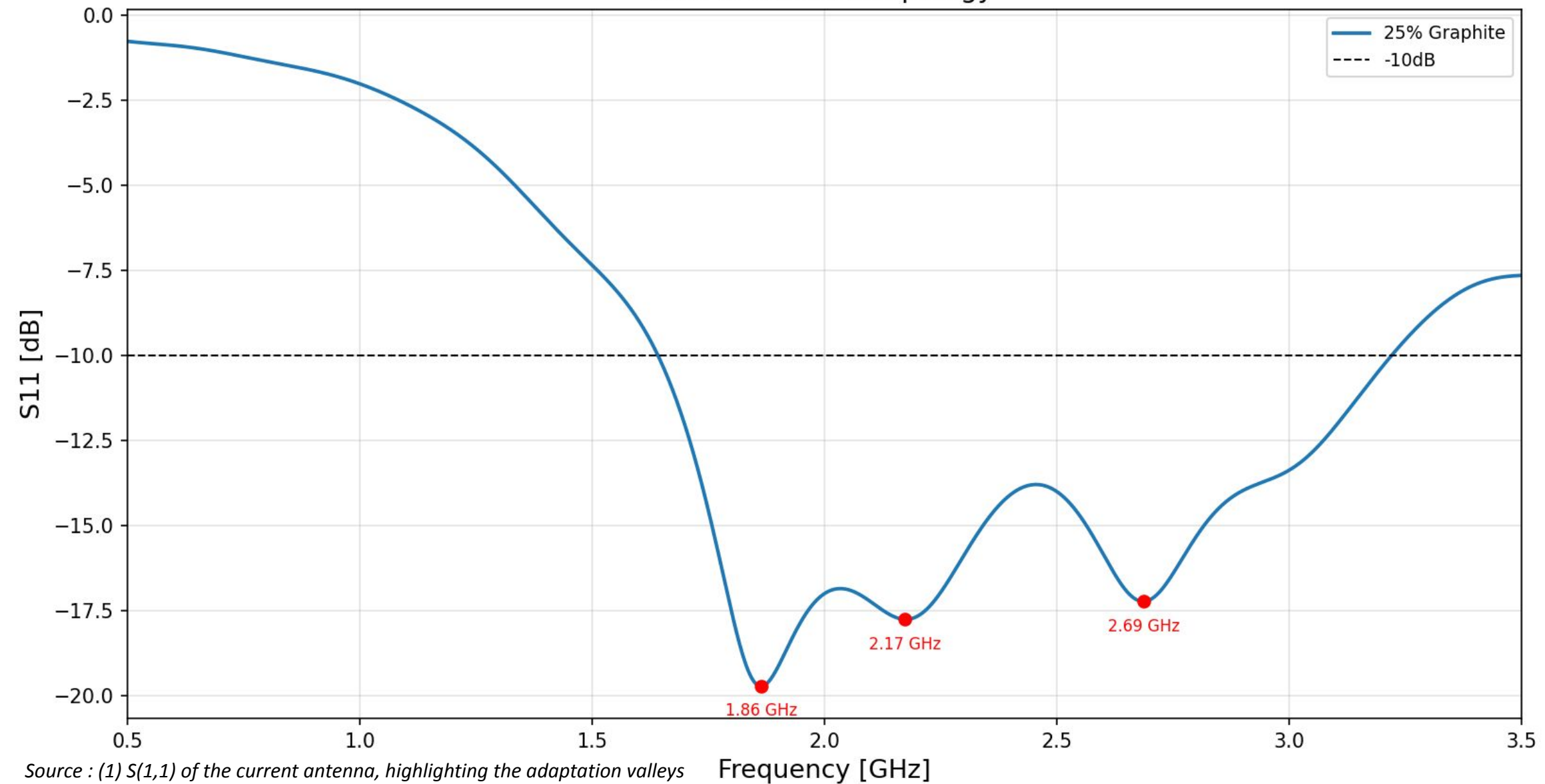
$$2\pi r \approx \frac{\lambda_g}{2}$$
$$r \approx \frac{\lambda_g}{4\pi}$$

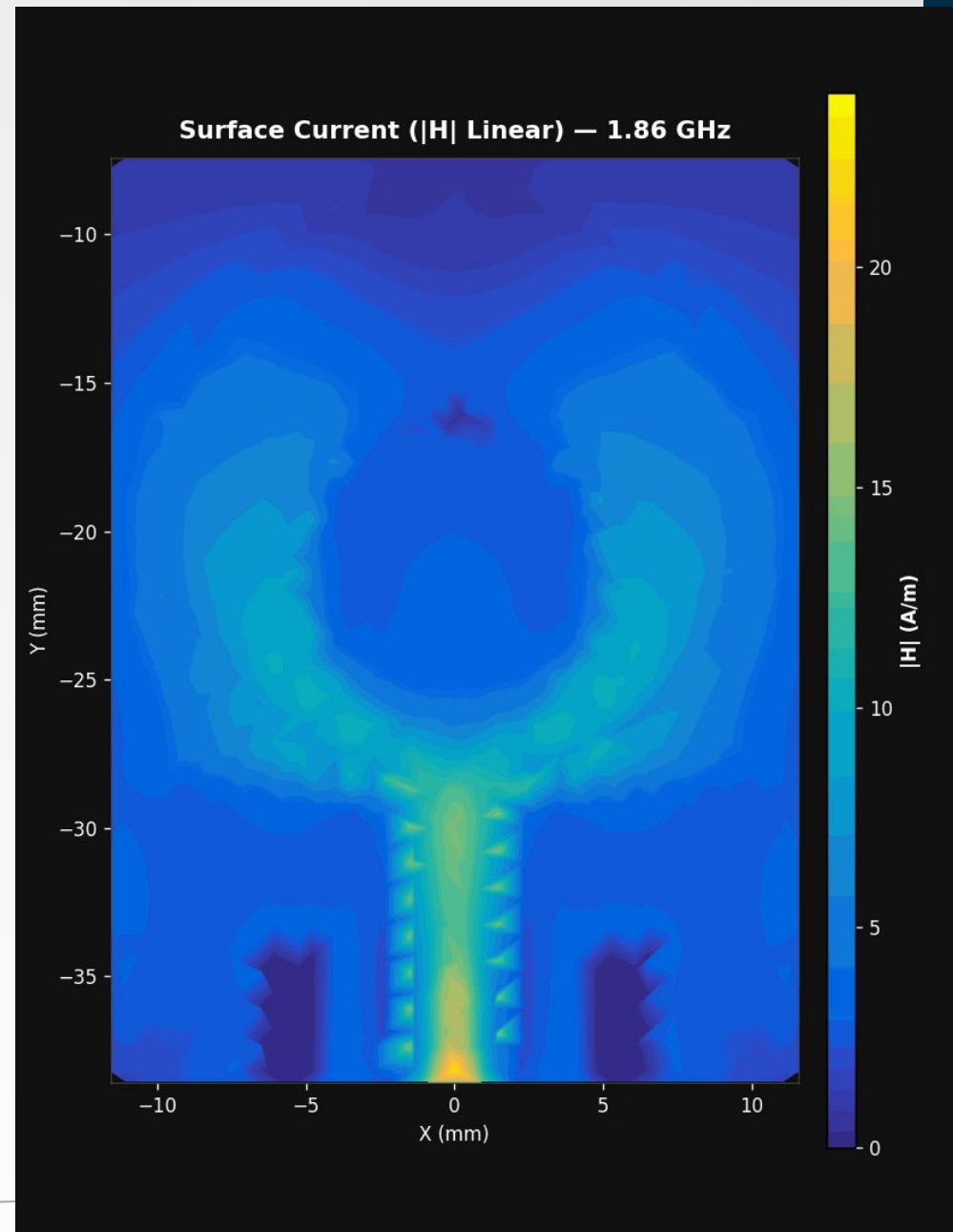
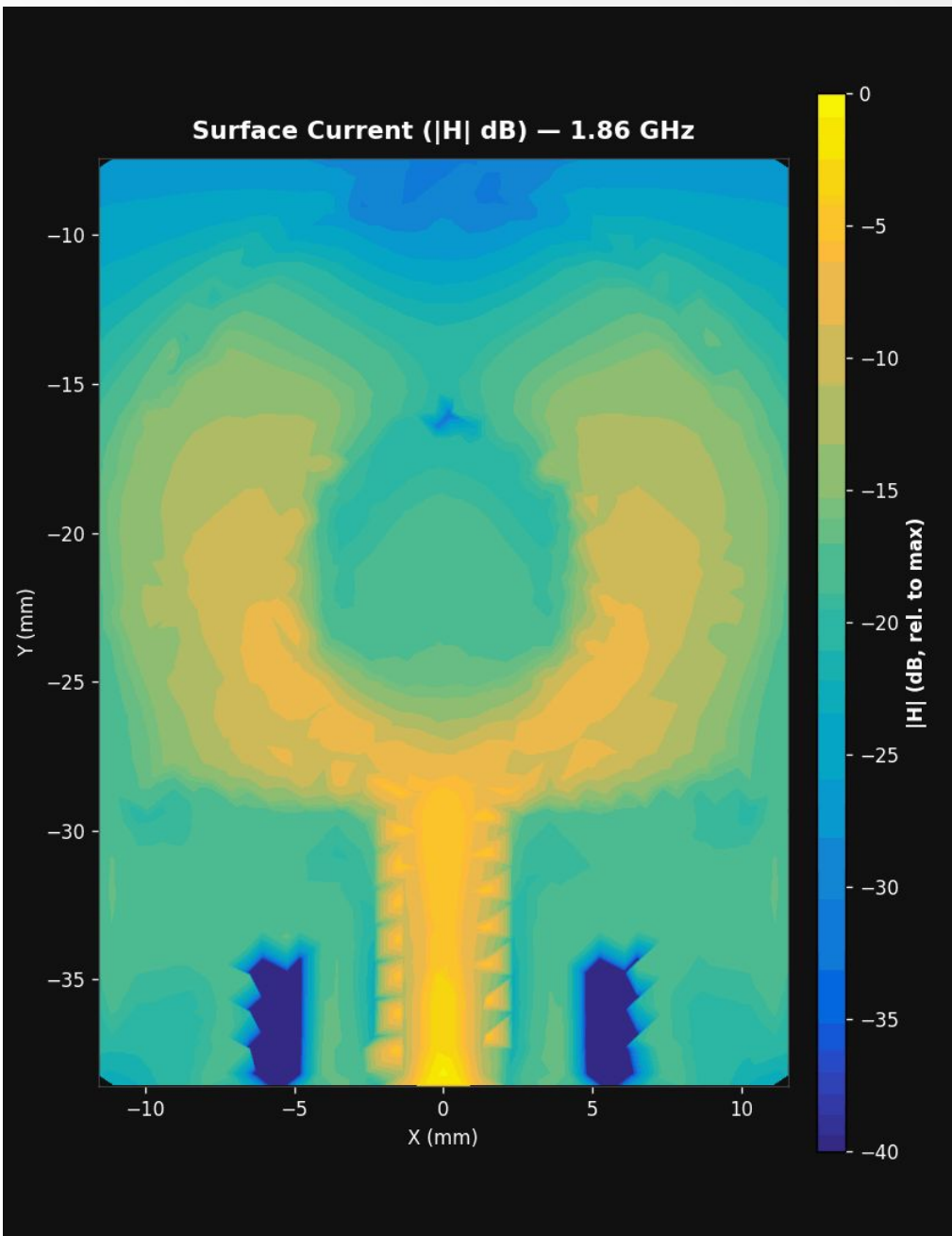
$$r \approx \frac{50.61}{4\pi} \approx 4.03 \text{ mm}$$

Choice of the **0.5** for the fundamental mode because of *size restriction* :

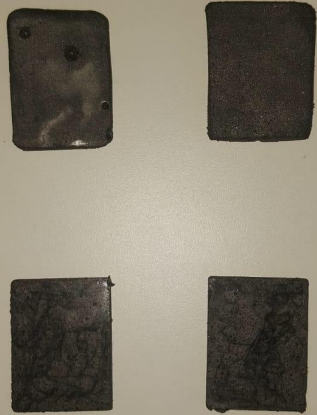
For $n = 1$ we get a mean radius of **8mm**, but the patch radius is **10mm**

S11 of the best topology

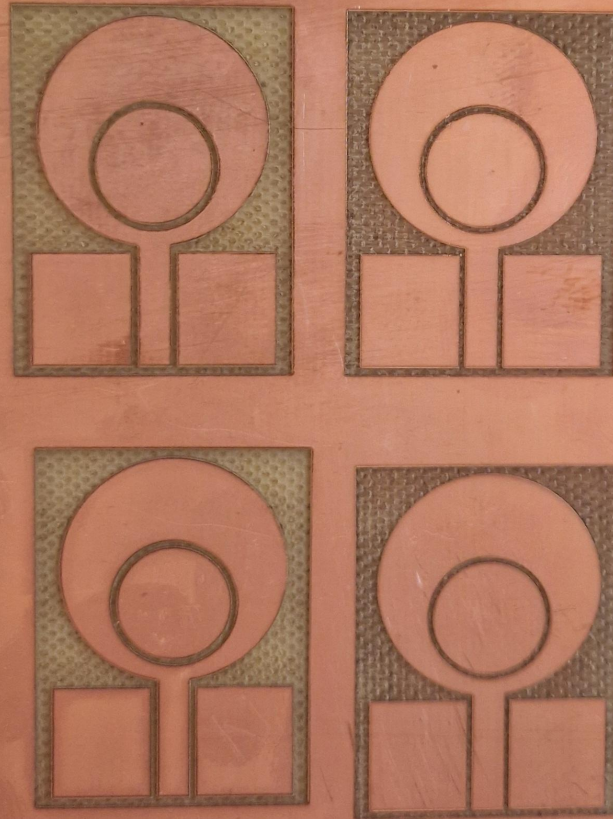




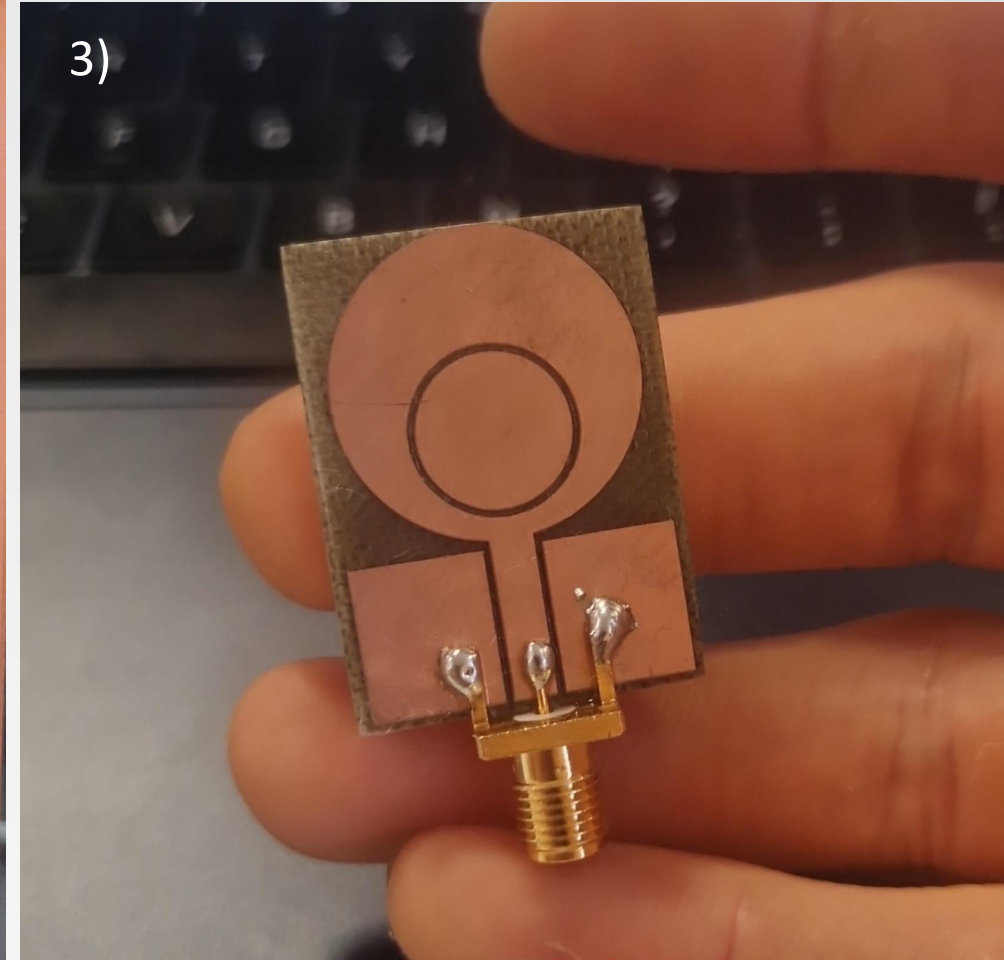
1)

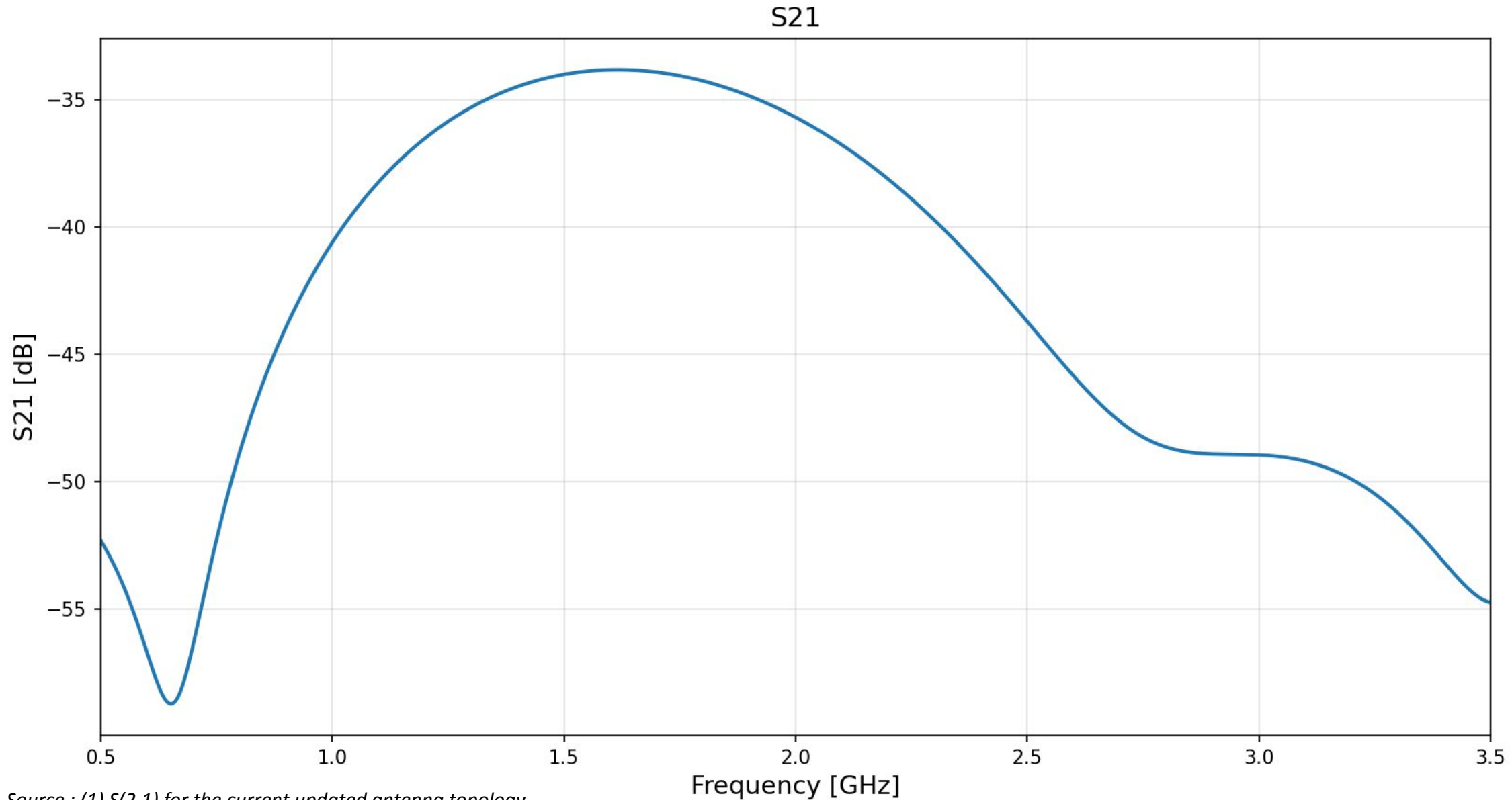


2)



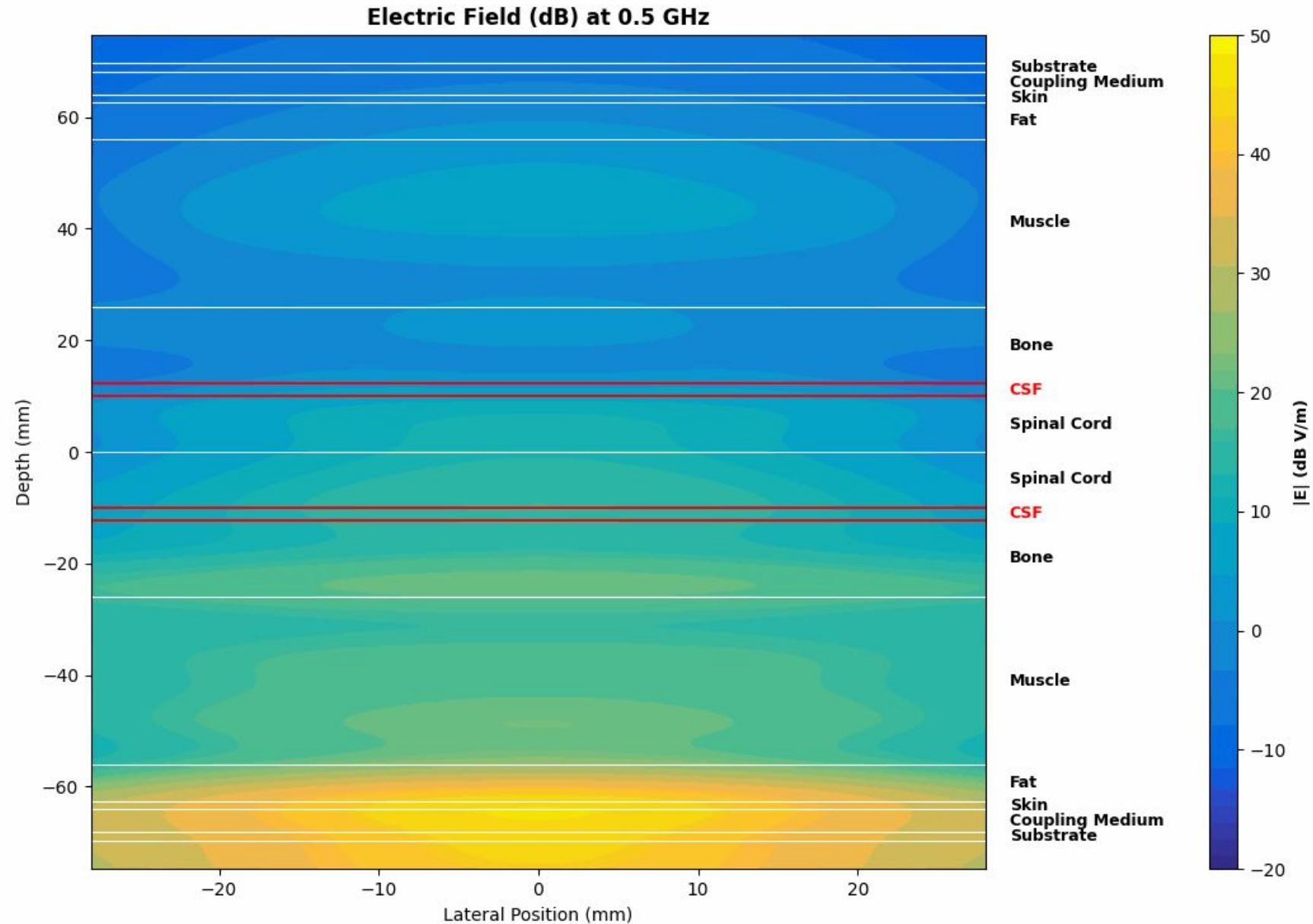
3)





Simulations

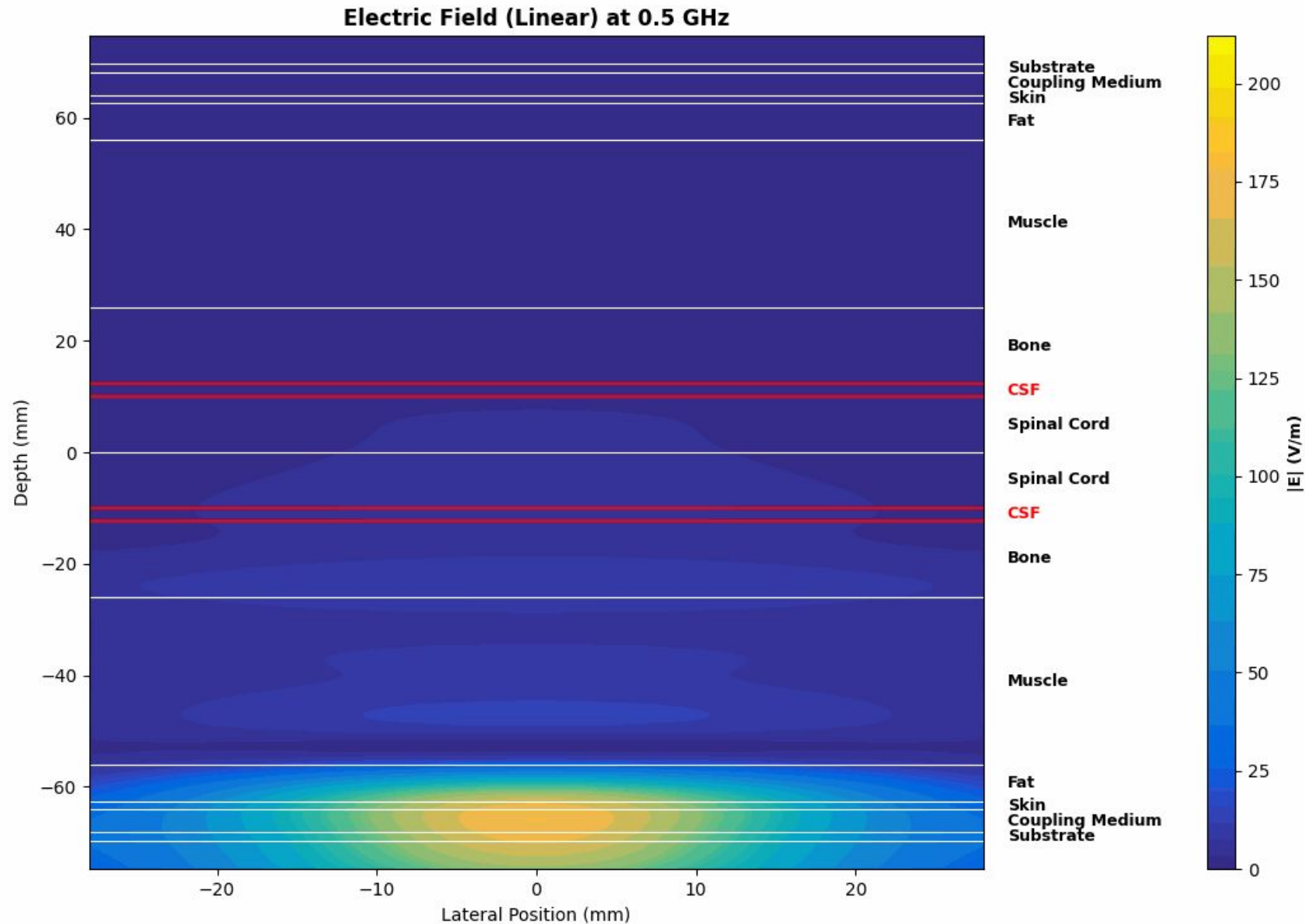
Electric Field (dB)



Source : Electric field visualisation in dB

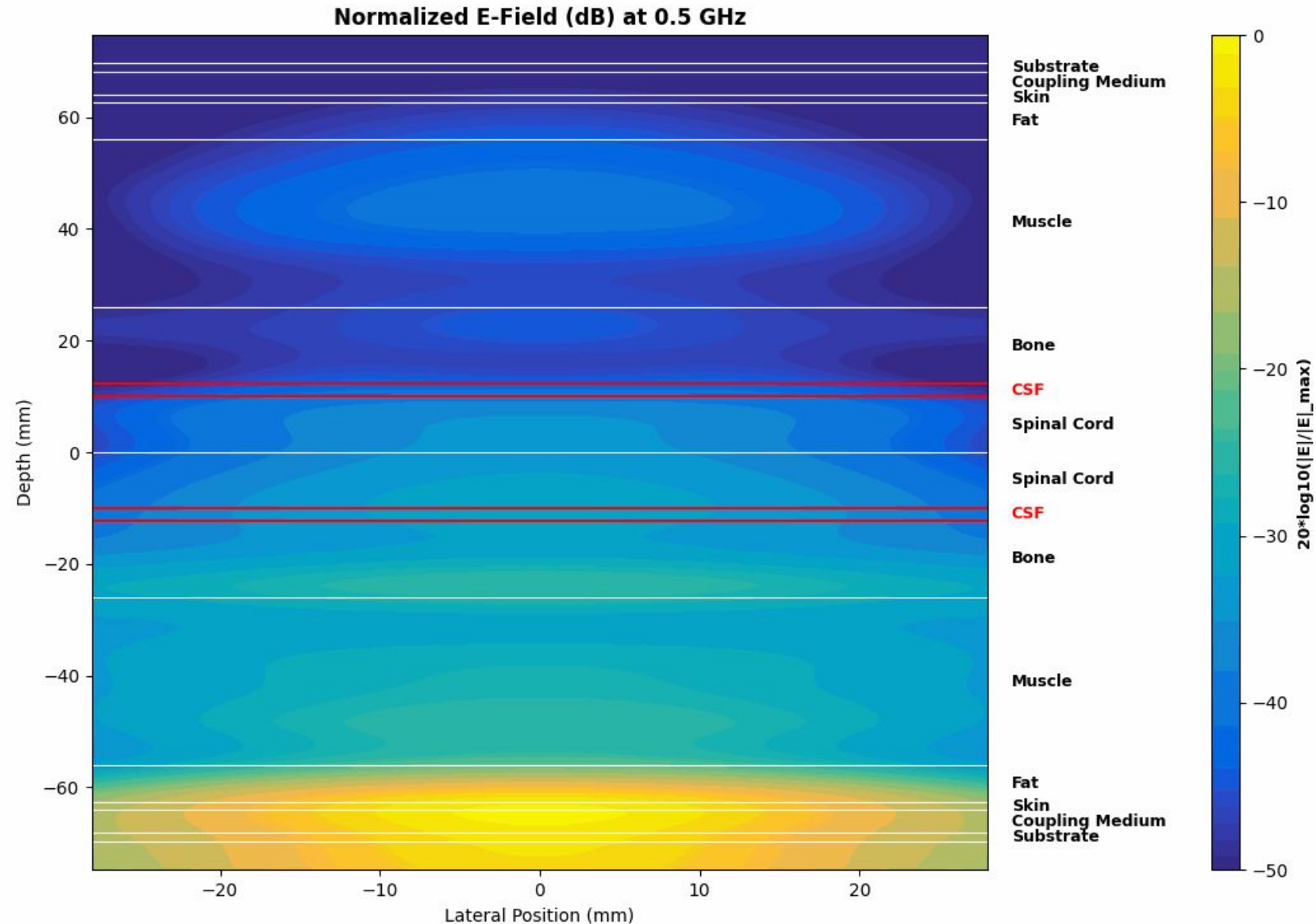
Simulations

E-field (linear)



Source: E-field visualisation in linear units

Simulations Normalized E-field (dB)

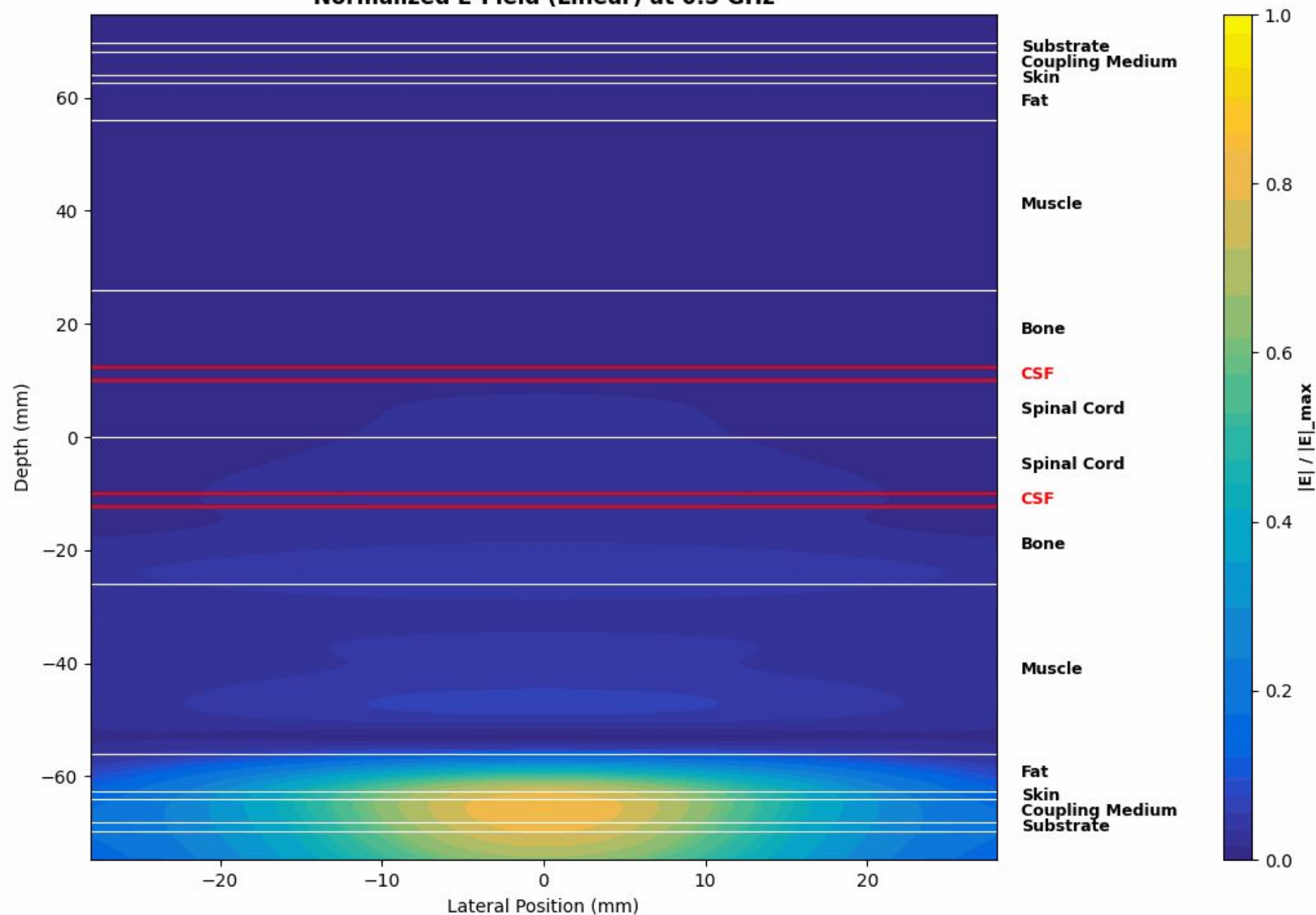


Source : E-field normalized by it's maximum value in dB

Simulations

Normalized E-field visualization

Normalized E-Field (Linear) at 0.5 GHz

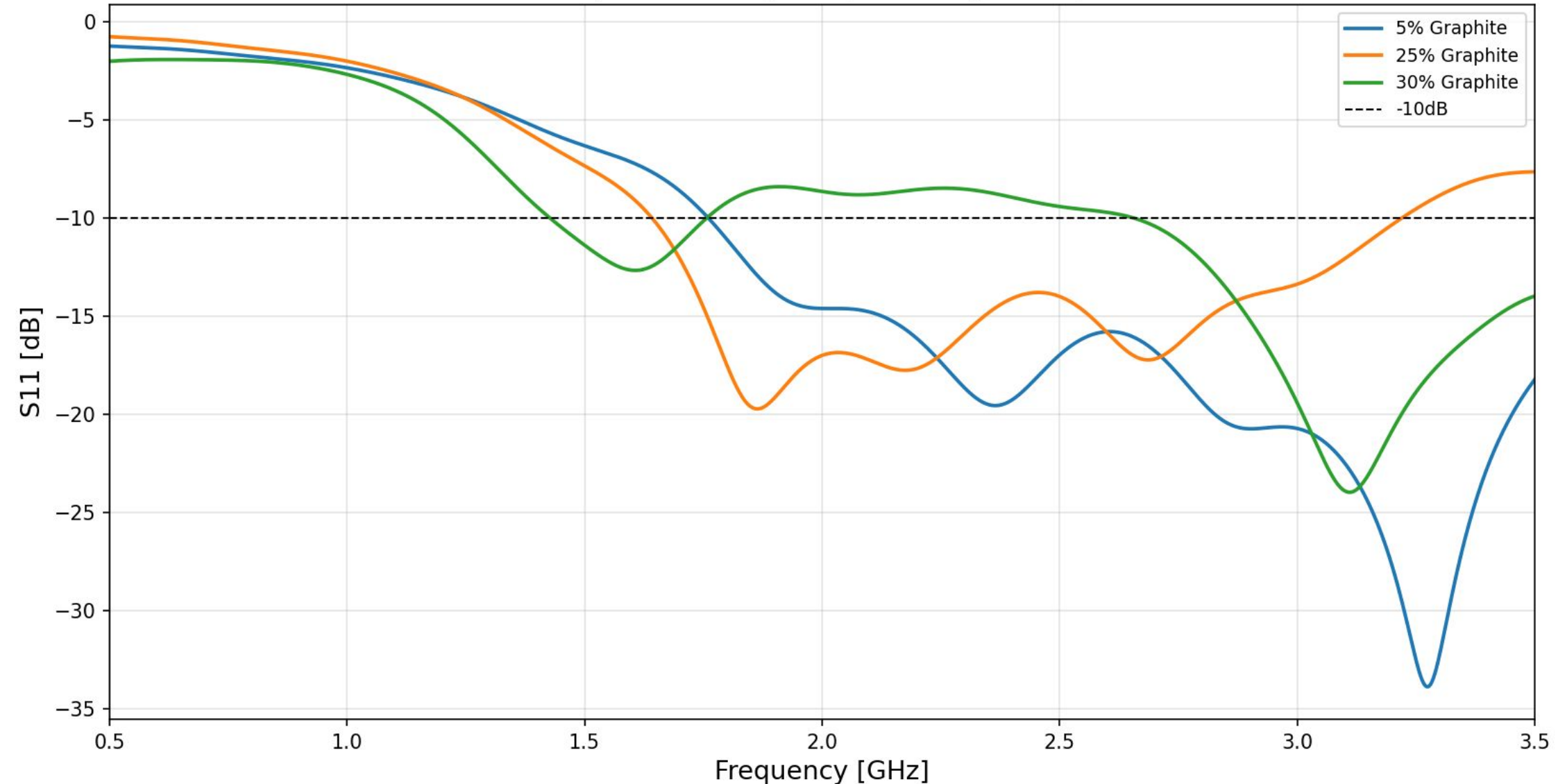


Source : E-field normalized by it's maximum value in linear units

Simulations

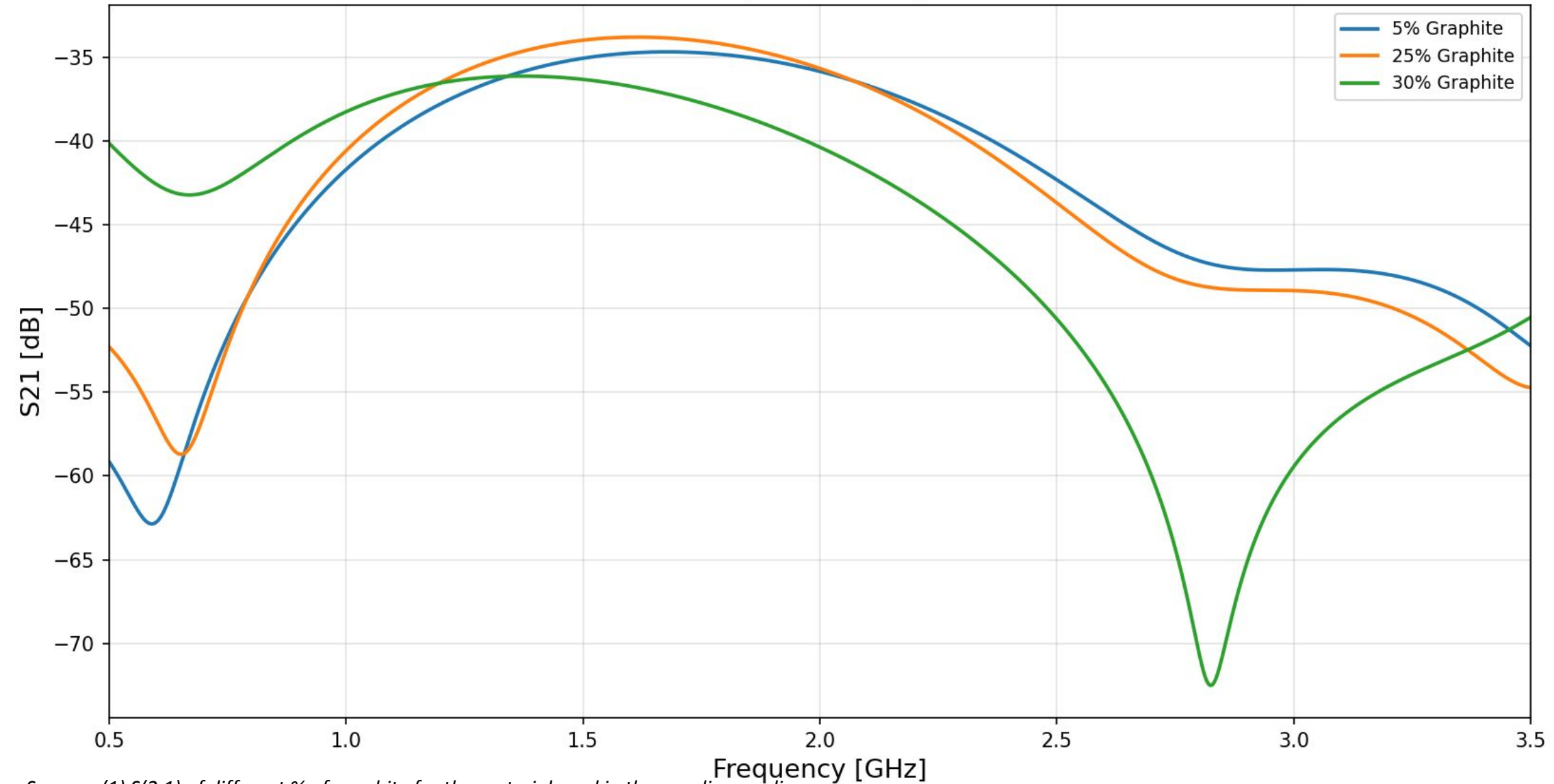
Changing the % of graphite

S11 Comparison: Graphite percentage comparison



Source : (1) $S(1,1)$ of different % of graphite for the material used in the coupling medium

S21 Comparison: Graphite %



Source : (1) $S(2,1)$ of different % of graphite for the material used in the coupling medium

Neck thickness

	NORMAL	AVERAGE	OVERWEIGHT
Study 1	11.05cm	12.3cm	>12.35cm
Study 2	11.09cm	11.3cm	>11.52cm

Maximum value : 13.35cm

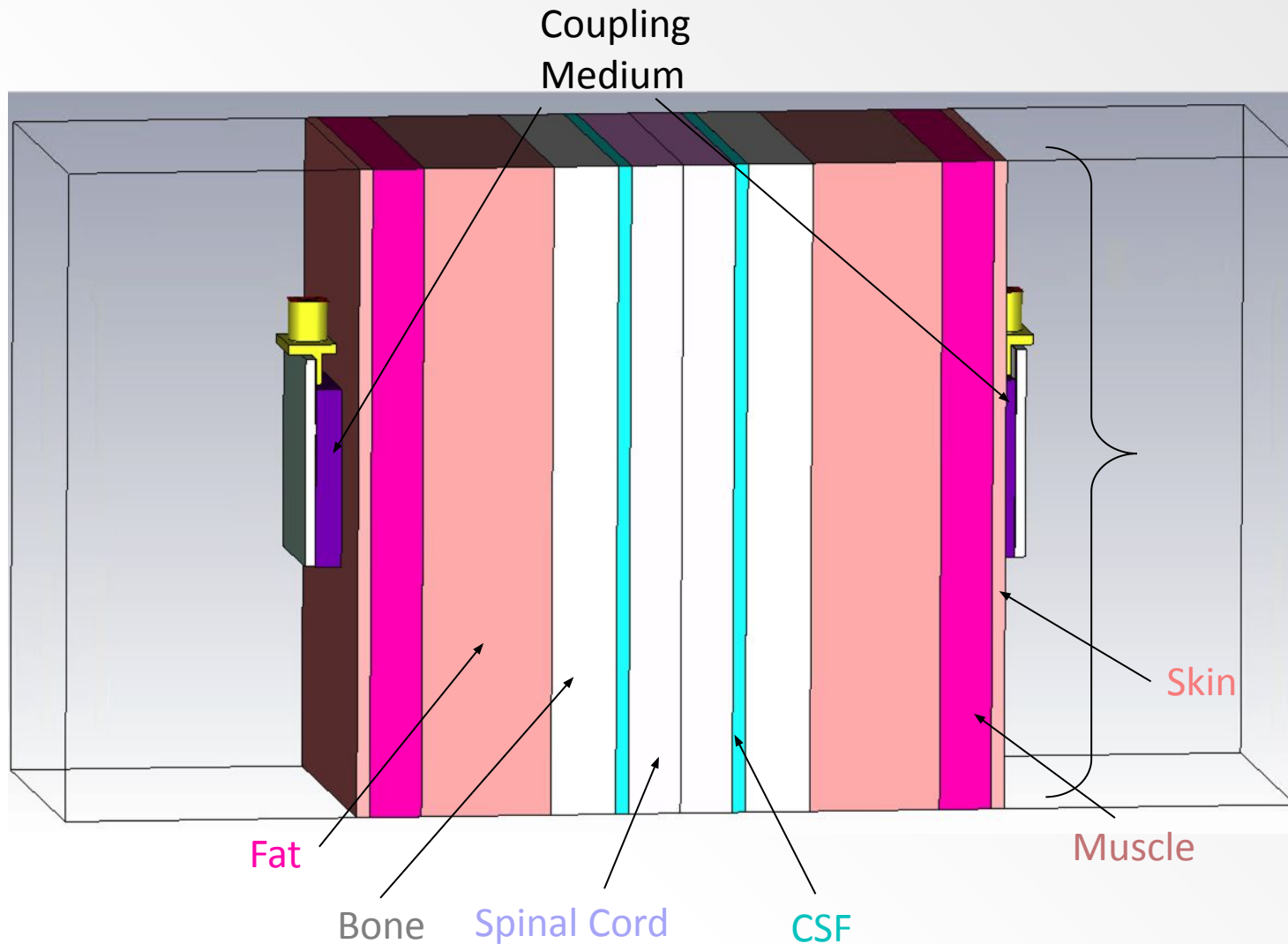
Current value: 12.3cm (average from (1))

- (1) **Study 1** : *Neck Circumference May Be a Better Alternative to Standard Anthropometric Measures*
- Healthy Hispanics (no *diabetes* or *cardiovascular* diseases)

- Study 2** : *Neck circumference as a useful marker of obesity: A comparison with body mass index and waist circumference*
- Pakistani dental students

MultiTissue block

Thicker models



Fat & Muscle * 1.15

LAYERS	THICKNESS	/ λ
Fat	0.67 -> 0.77	0.025
Muscle	3.0 -> 3.45 cm	0.11

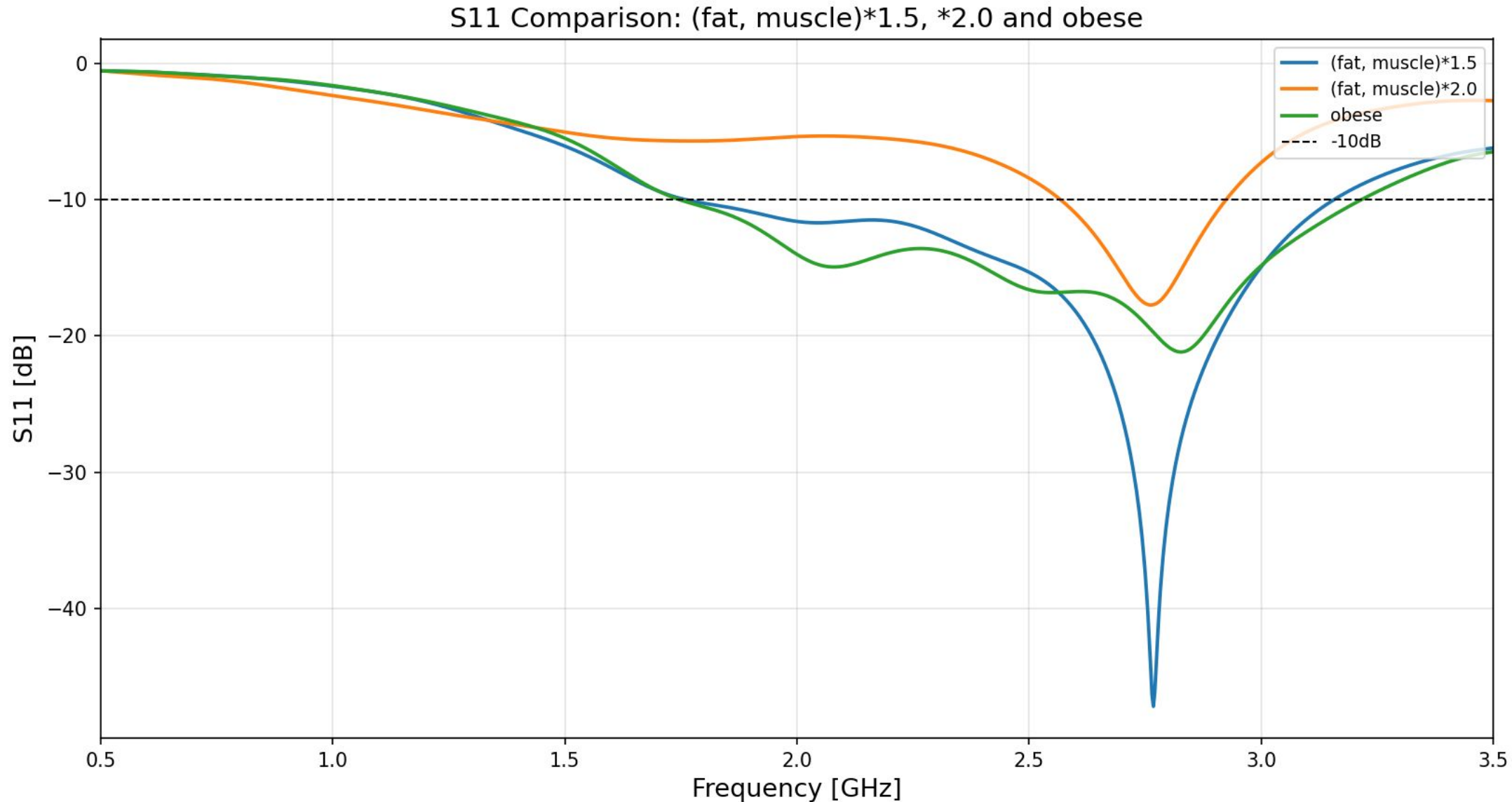
Fat & Muscle * 1.5

LAYERS	THICKNESS	/ λ
Fat	0.67 -> 1.005cm	0.033
Muscle	3.0 -> 4.5 cm	0.15

Fat & Muscle * 2.0

LAYERS	THICKNESS	/ λ
Fat	0.67 -> 1.34cm	0.045
Muscle	3.0 -> 6.0 cm	0.2

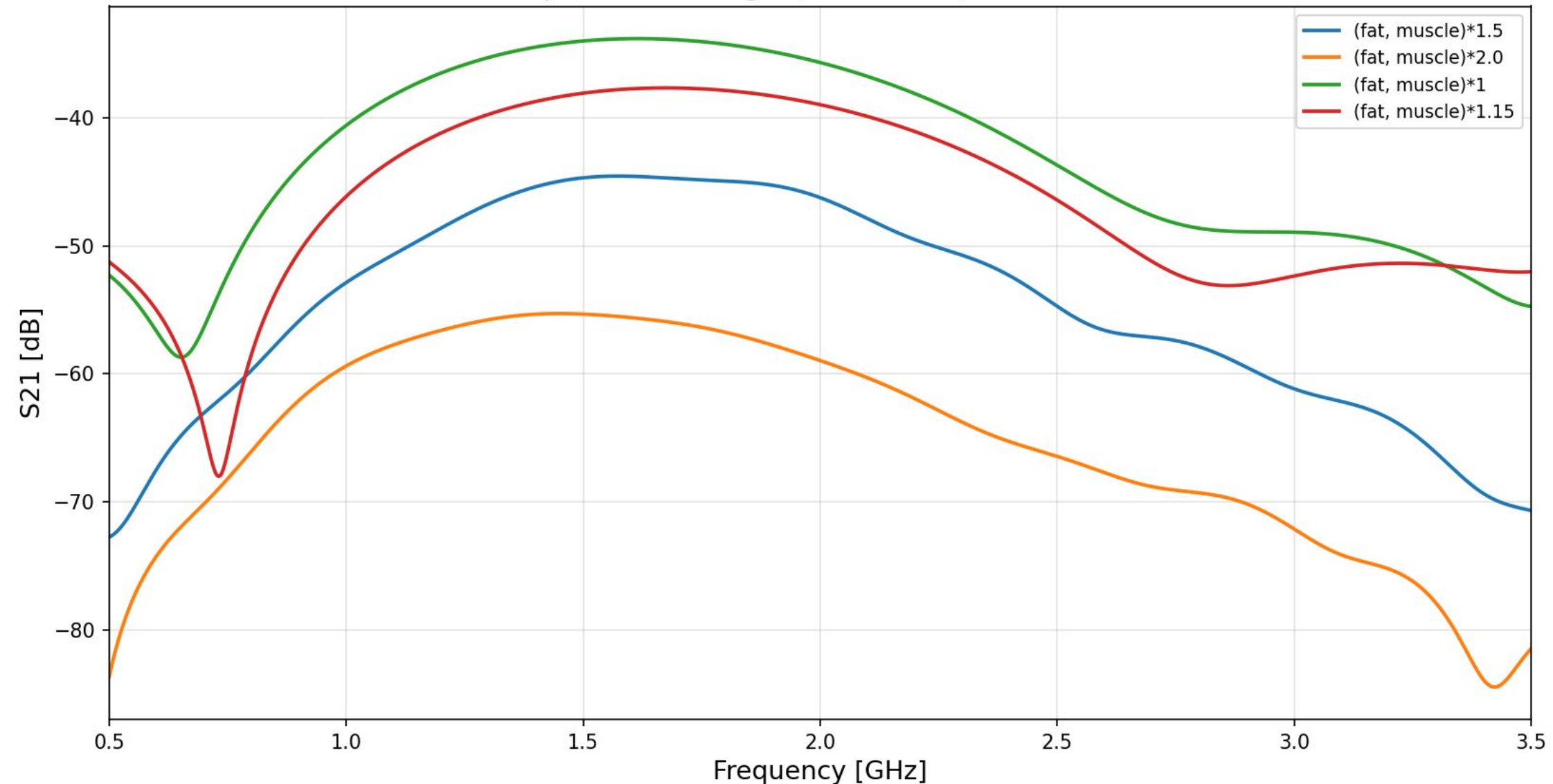
Simulations Thicker necks



Source : (1) $S(1,1)$ comparison of the first thicker model (*1.5), the second one (*2.0) and realistic obese model

Simulations Thicker necks

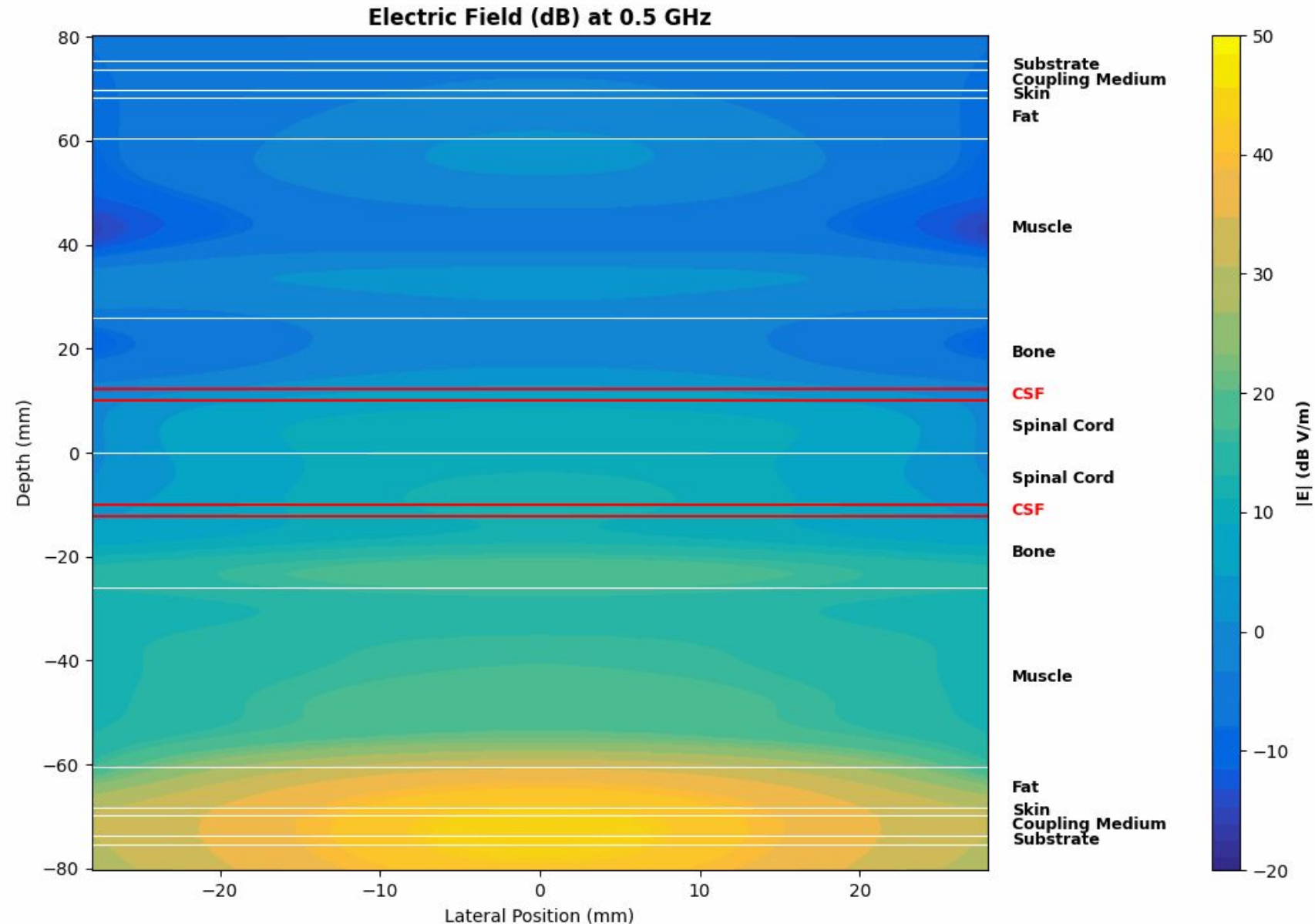
S21 Comparison: Average, (fat, muscle)*1.15, 1.5 and 2.0



Source : (1) $S(2,1)$ comparison of the first thicker model ($\times 1.5$), the second one ($\times 2.0$) and realistic obese model

Simulations

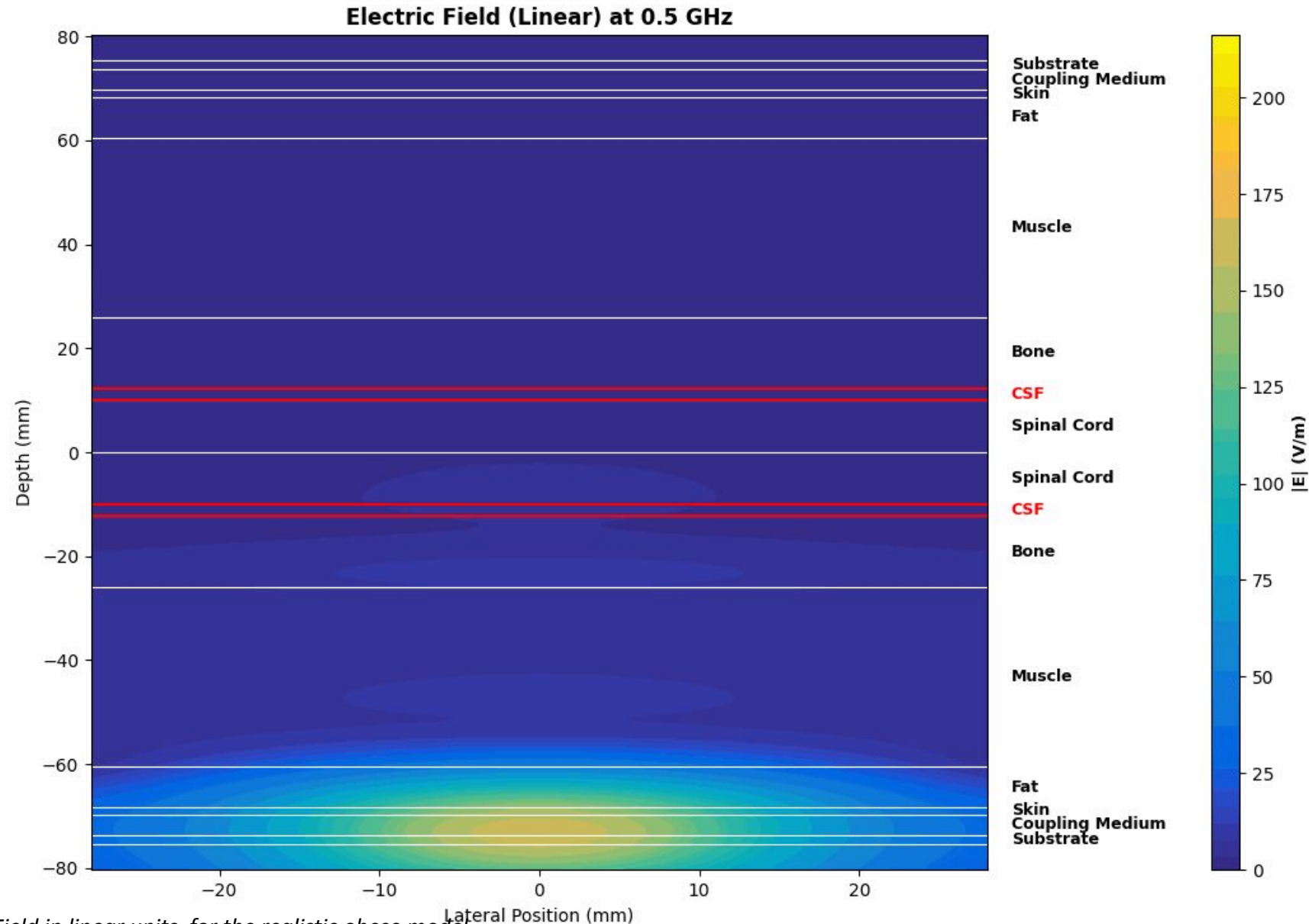
Electric Field (dB) realistic obese



Source : Electric Field in dB for the realistic obese model

Simulations

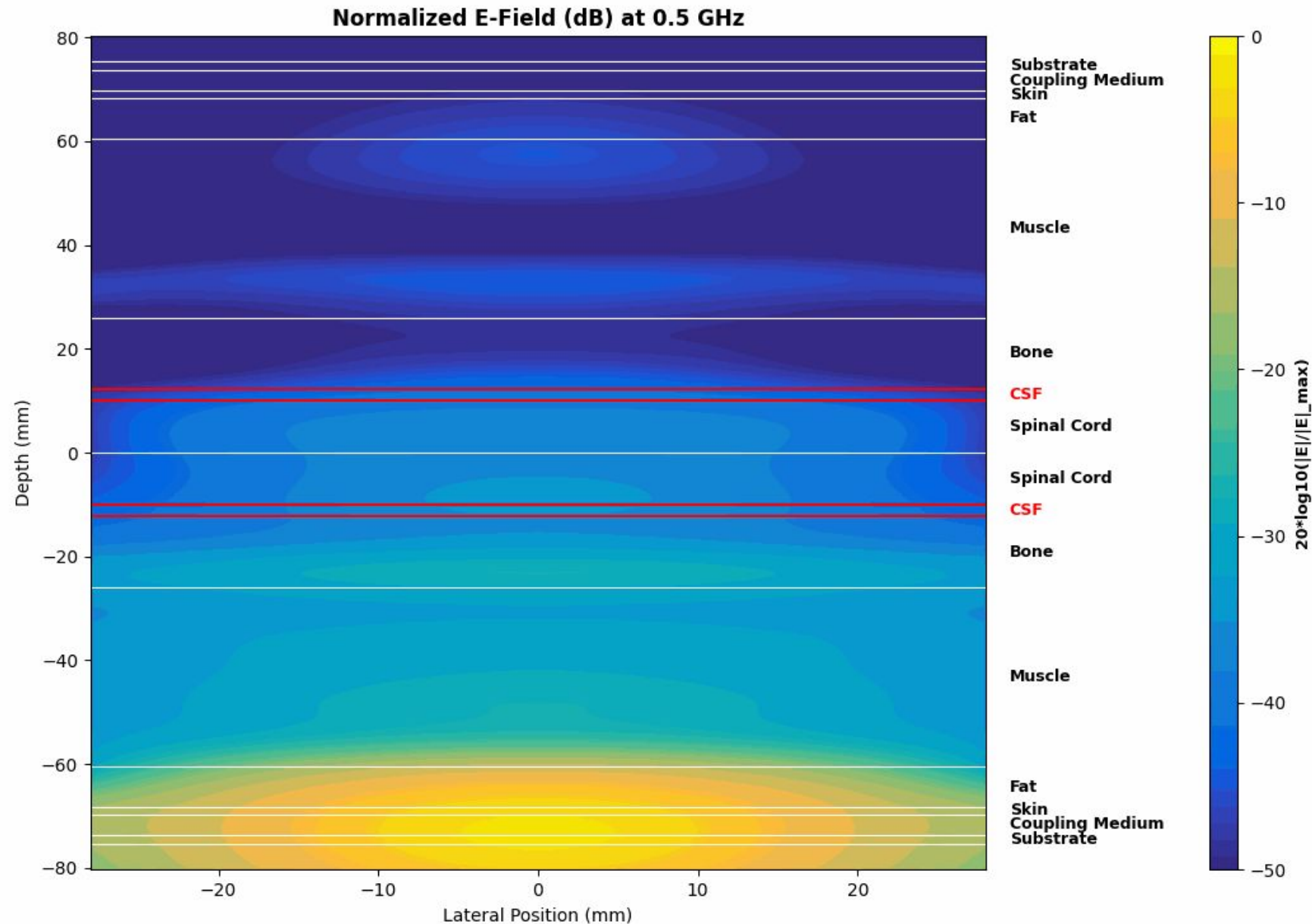
Electric Field (linear) realistic obese



Source : Electric Field in linear units for the realistic obese model

Simulations

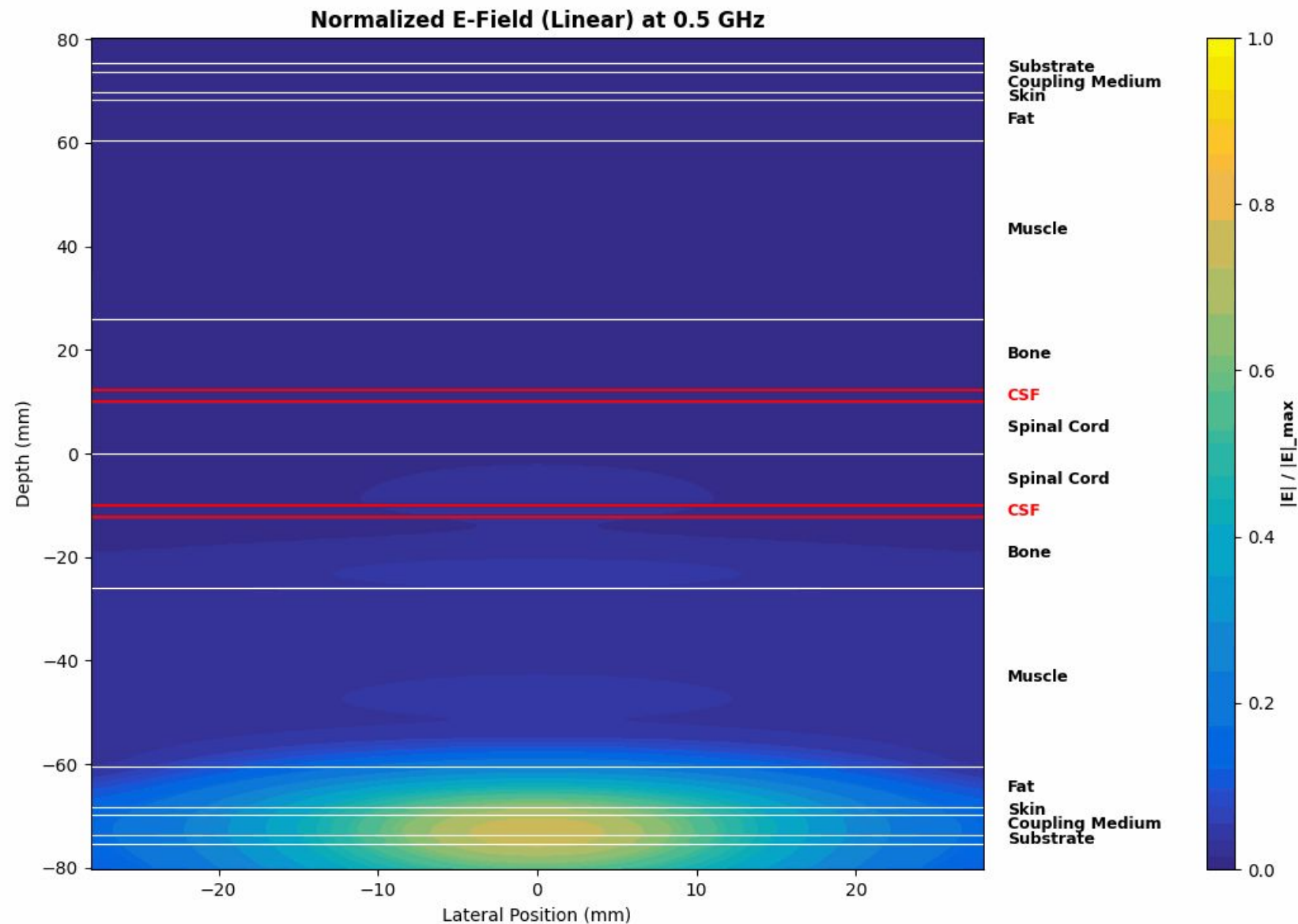
Normalised Electric Field (dB) realistic obese



Source : Normalised Electric Field in dB for the realistic obese

Simulations

Normalised Electric Field (linear) realistic obese



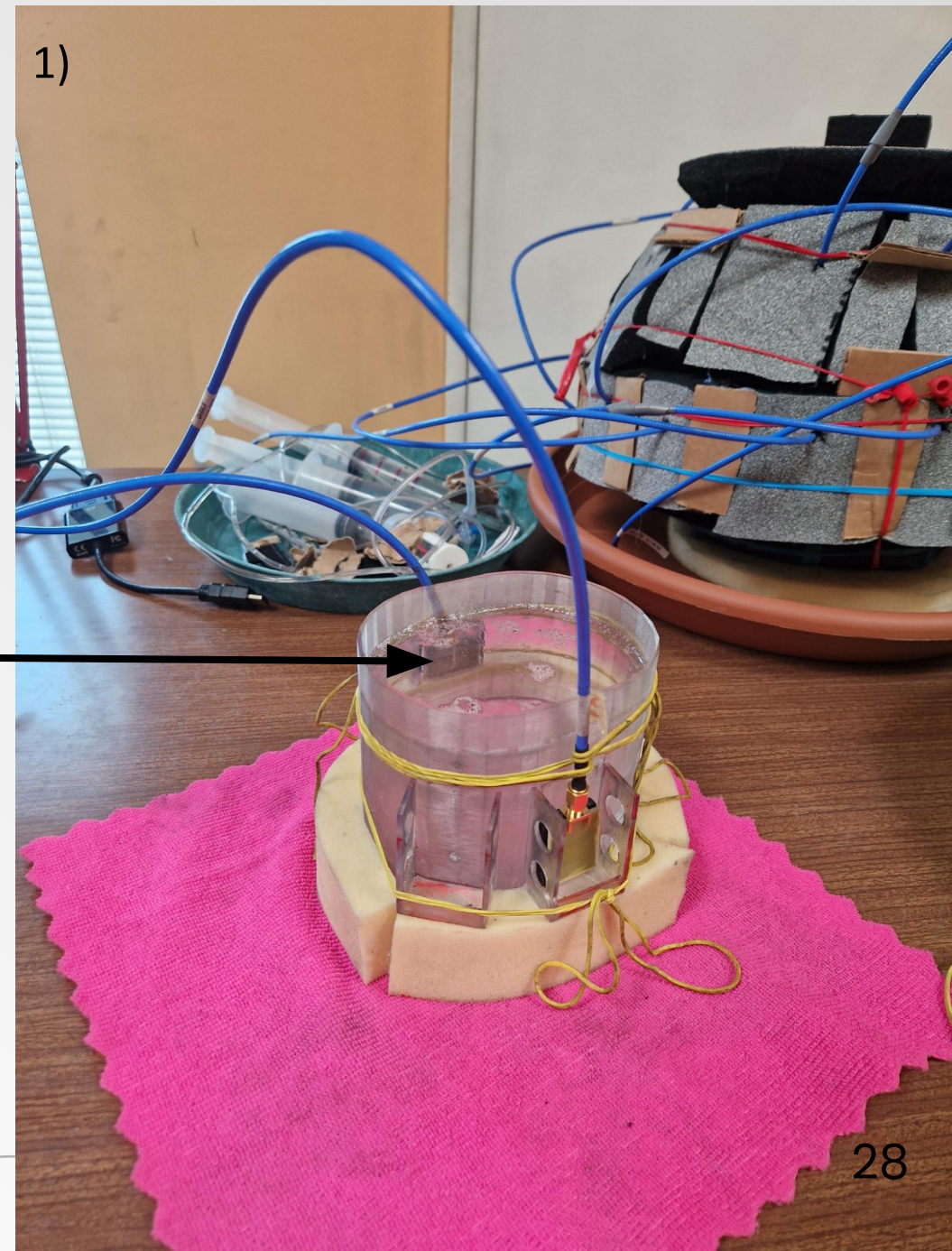
Source : Normalised Electric Field in linear units for the realistic obese model

Experiments

Average Neck

Liquid which mimics the neck's dielectric properties made of NaCl, TritonX100 and water. Proportions computed by:

- Calculating the weight of every tissue layer ($w_i = \text{tissue_layer_thickness} / \text{total_thickness}$)
- Then using a model (2) to compute the mass of NaCl and TritonX100 for each layer (M_{NaCl} , M_{TX100})
- Calculating the weighted average for each of those ingredient by multiplying each weight with the mass of the ingredients ($M_{\text{NaCl}} * w_i$, $M_{\text{TX100}} * w_i$)
- Summing all of the masses ($M_{\text{NaCl}} = \text{sum}(M_{\text{NaCl}})$, $\text{sum}(M_{\text{TX100}})$)



Experiments

Average Neck

The *antennas* were made in LED by using a *LASER engraver*, with one batch of two antennas made by passing the LASER 5 times (**A5**) and the other one 6 times (**A6**)

On each experiment the *1st port* corresponds to **A6** and the *2nd* to **A5**

The *matching mediums* were made in the DET lab with a *two part rubber mixture* and *graphite*, one with a **5%** (by mass) of *graphite* the other with **20%**. Then poured into a 3D printed *PLA* mould. A **25%** and a **30%** mixture were attempted, but the rubber did not cure properly.

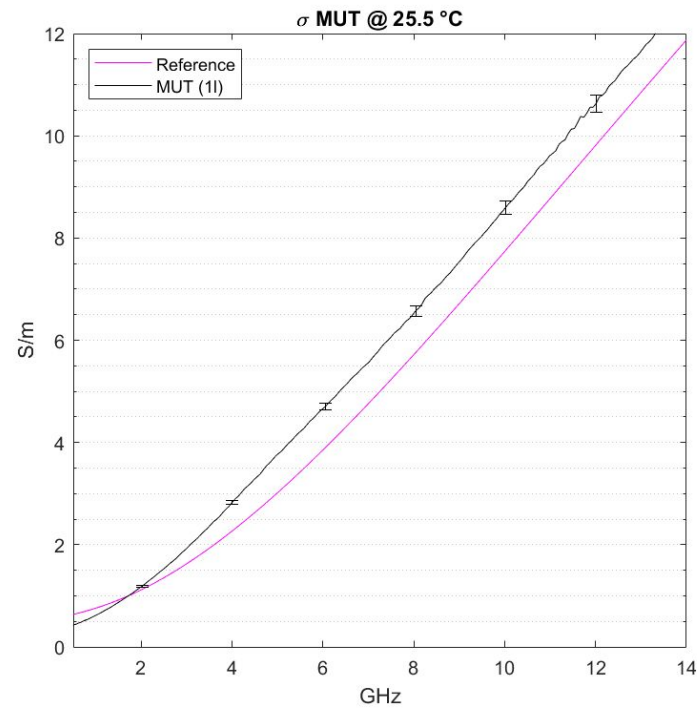
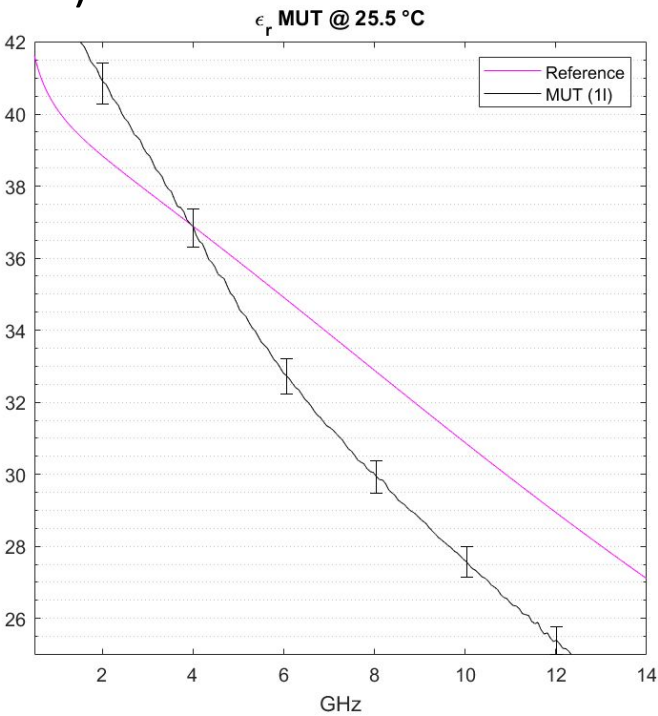
The bench is composed of a *VNA*, *phase stable coaxial cables*, *SLIM-4 probe*
A **1000 kHz** *IF filter* was chosen on the *VNA software*.

Then the simulations files were exported as *touchstones* files and plotted using *Python3.0*

Experiments

Average Neck

1)

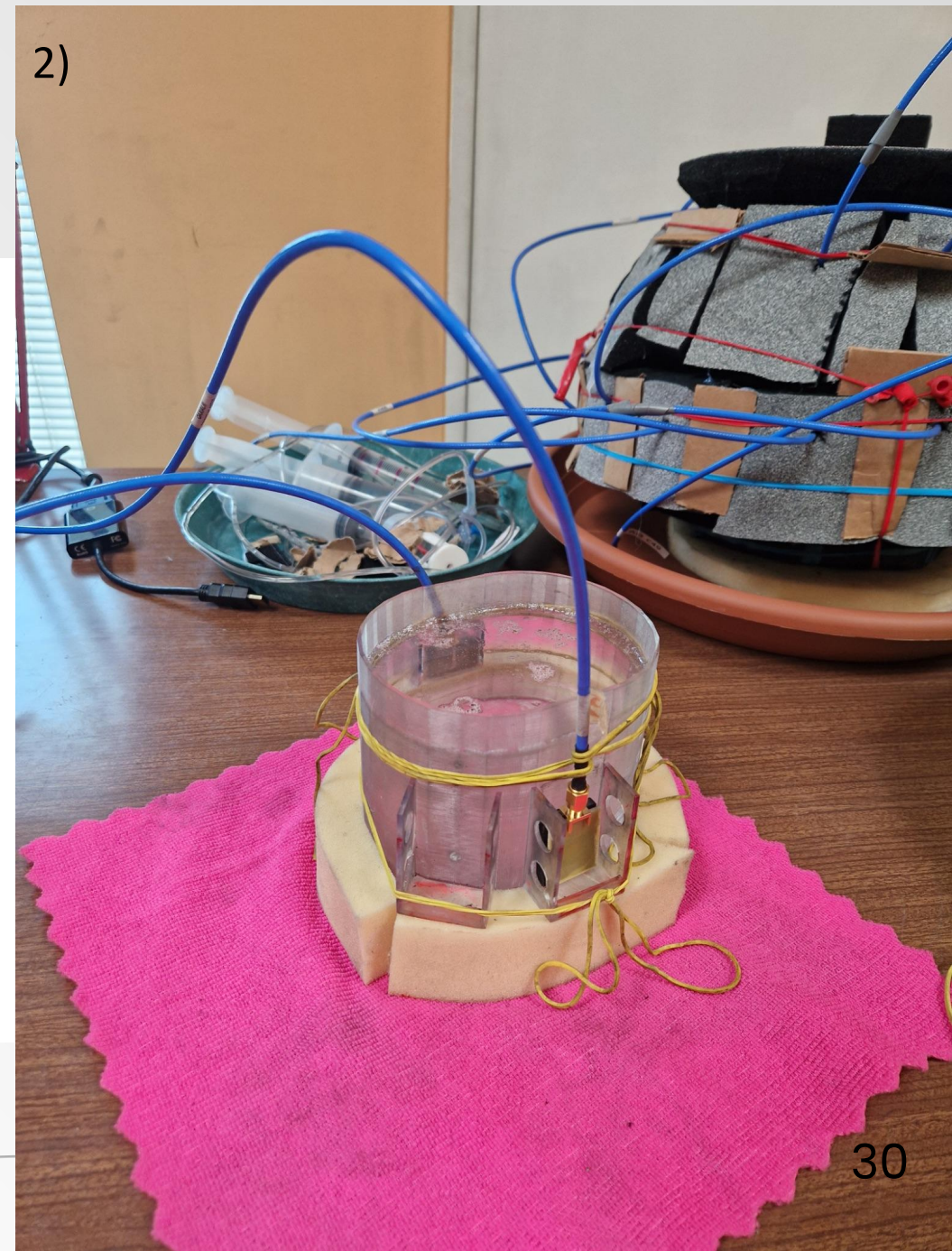


Reference -> **weighted average** of the neck model dielectric's properties

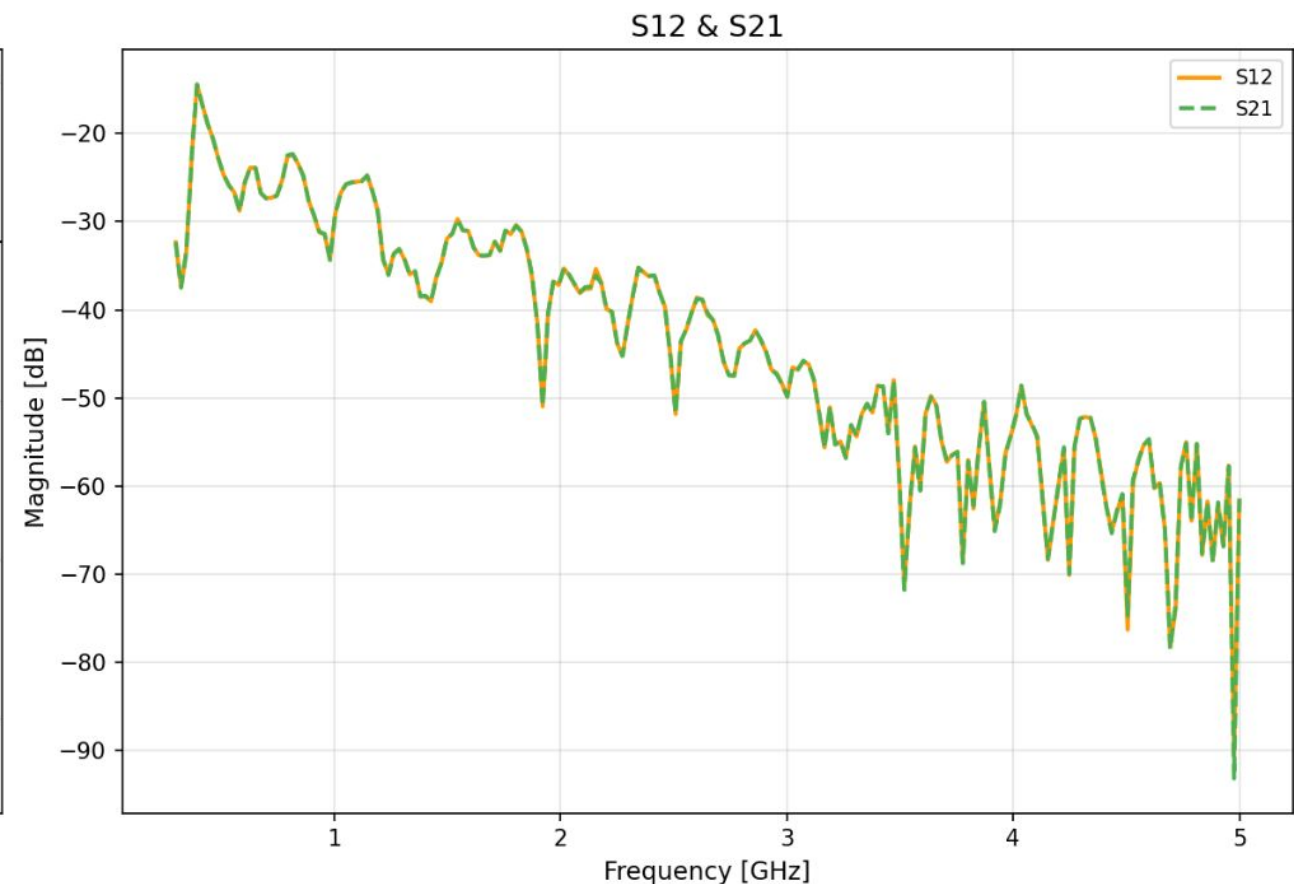
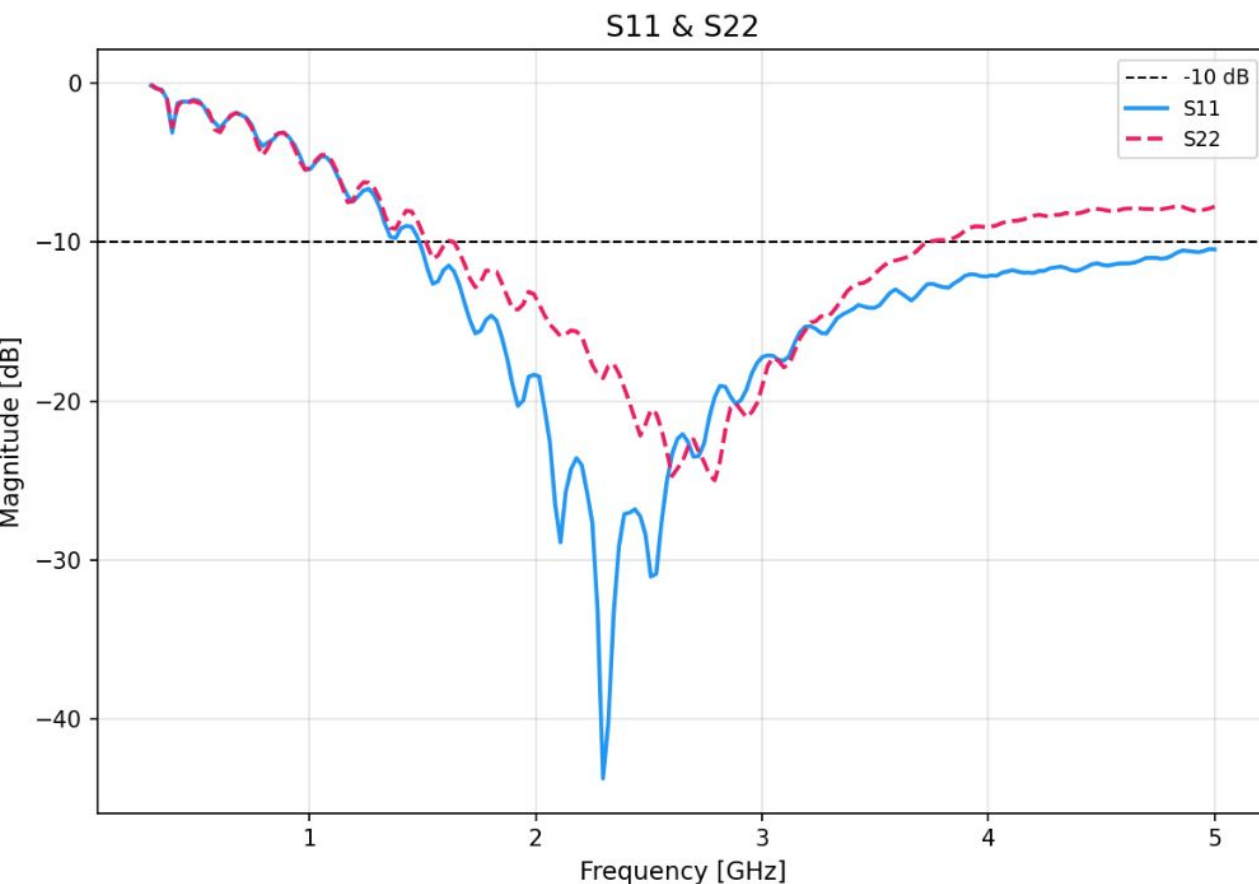
MUT -> Material Under Test

Error bars corresponding to the **min** and **max**

2)

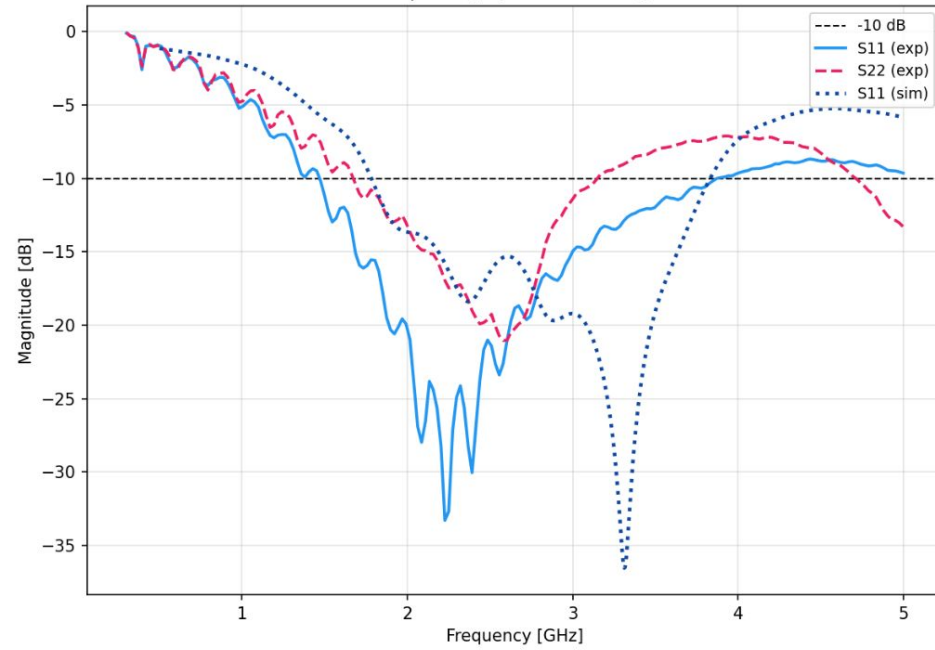


NeckPhantom — no matching medium

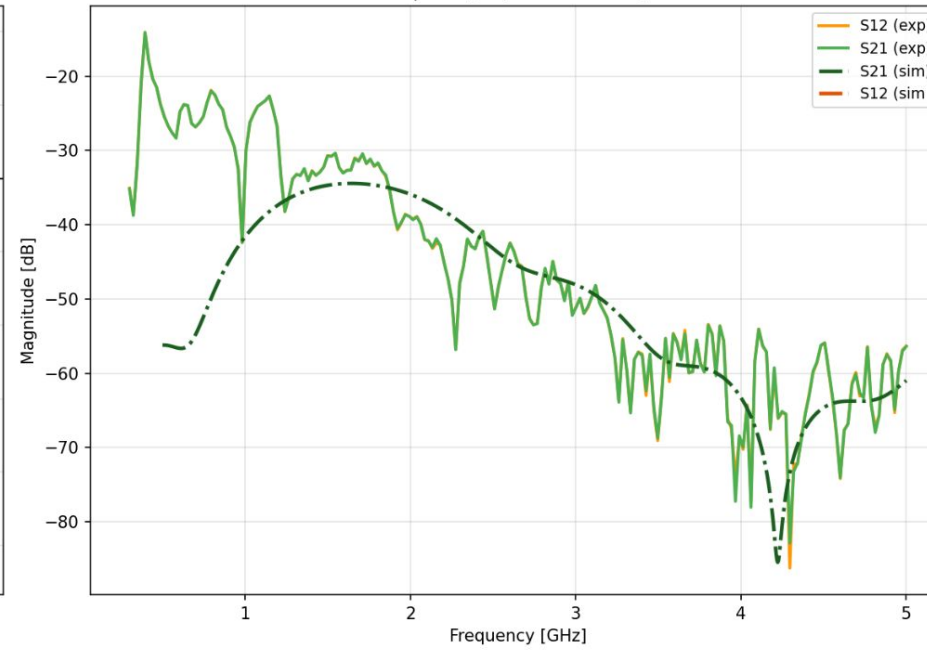


NeckPhantom — Graphite matching media comparison

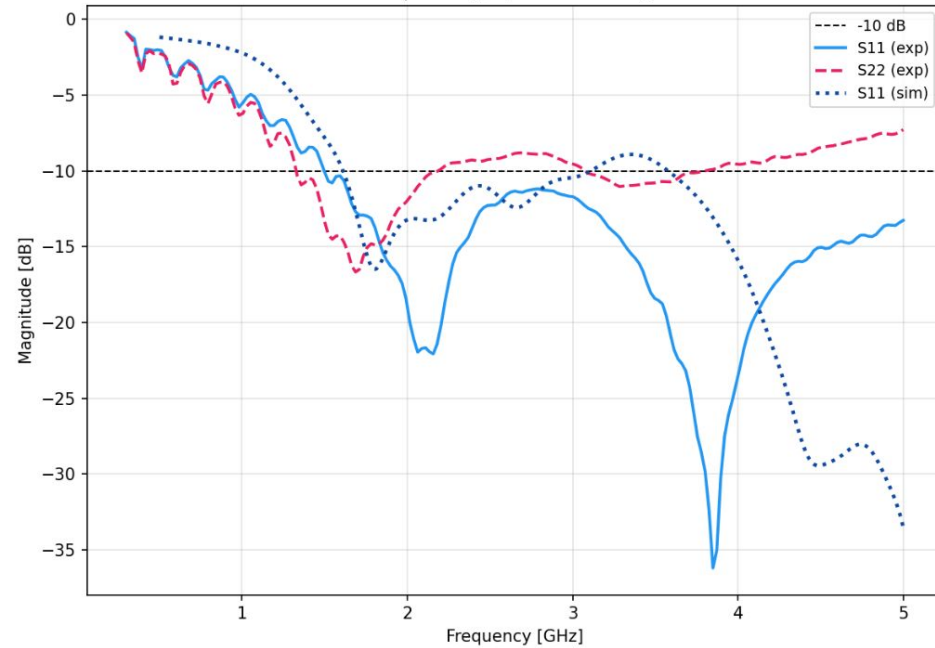
S11 & S22 | 5% graphite matching medium



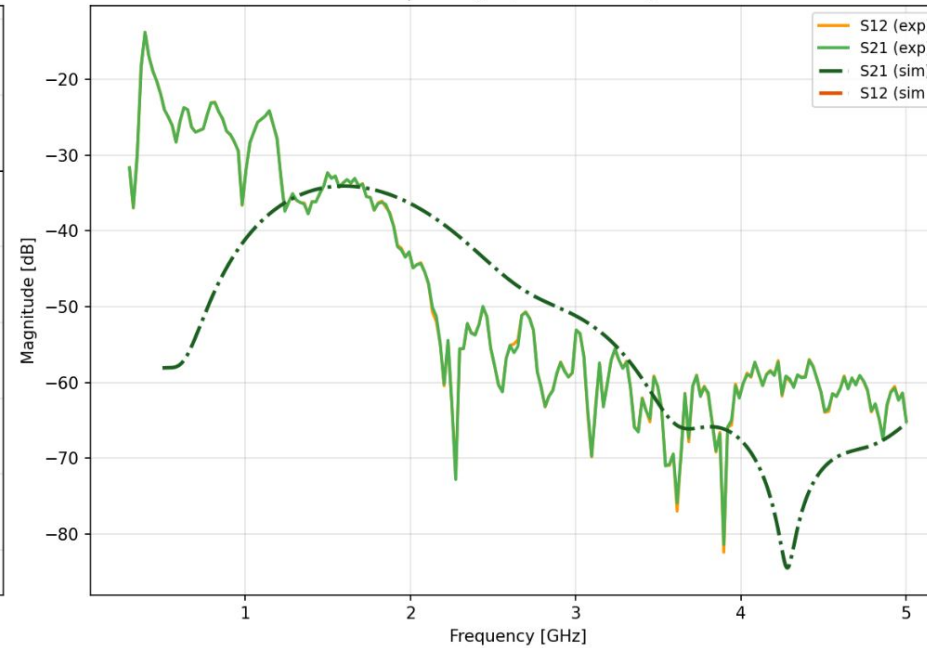
S12 & S21 | 5% graphite matching medium



S11 & S22 | 20% graphite matching medium

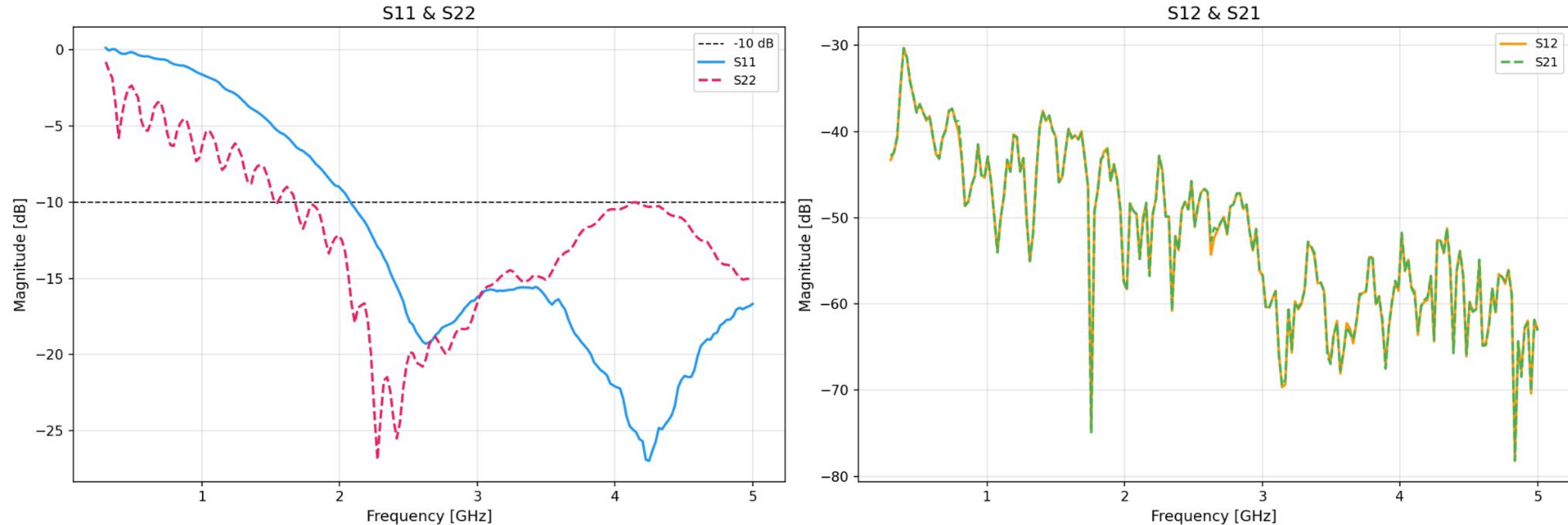


S12 & S21 | 20% graphite matching medium



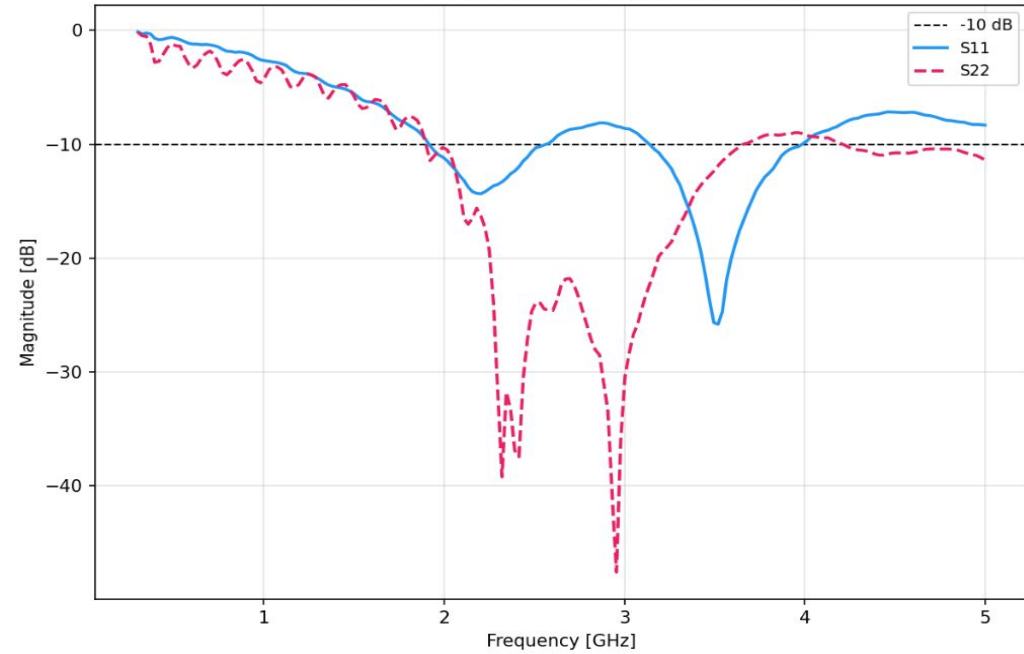
Source : Reflection and transmission parameters for the experiment on the Neck Phantom with a coupling medium layer made of either 5 or 20 % graphite

headPhantom — no matching medium

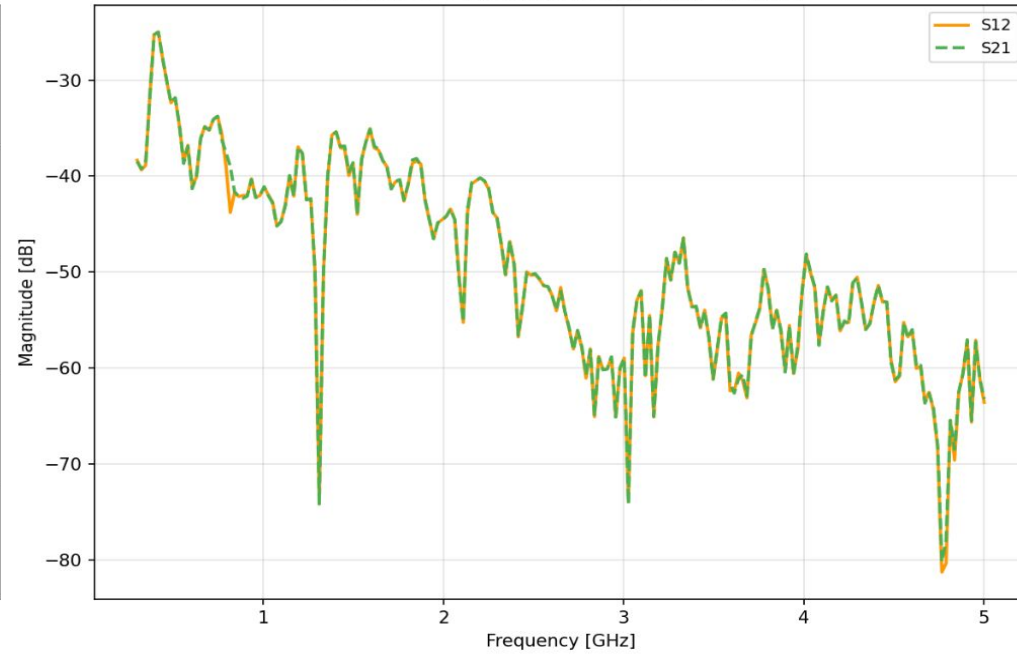


headPhantom — Graphite matching media comparison

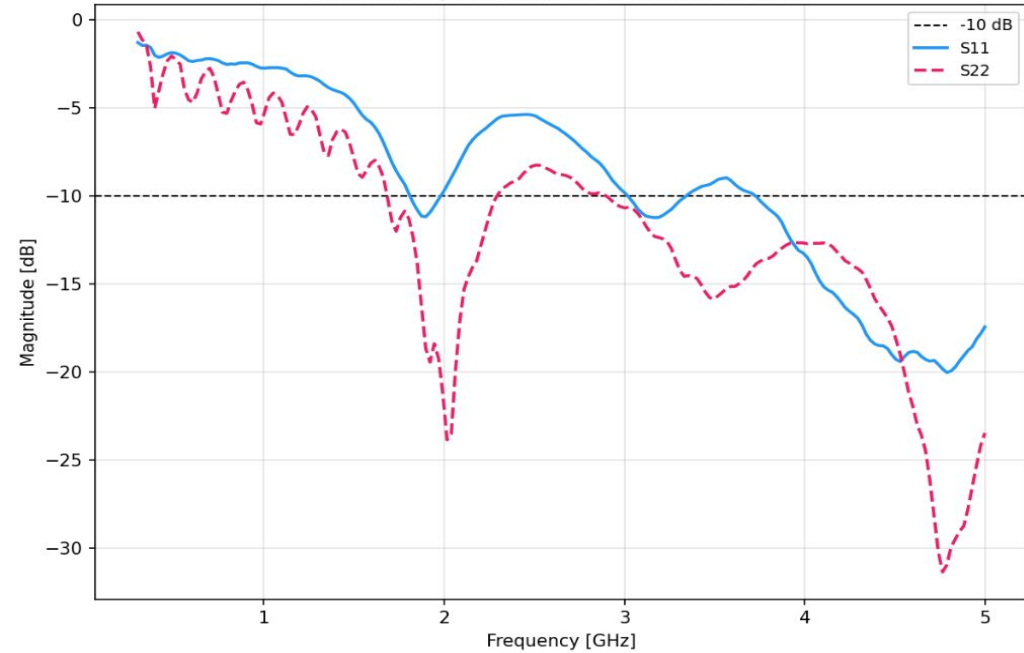
S11 & S22 | 5% graphite matching medium



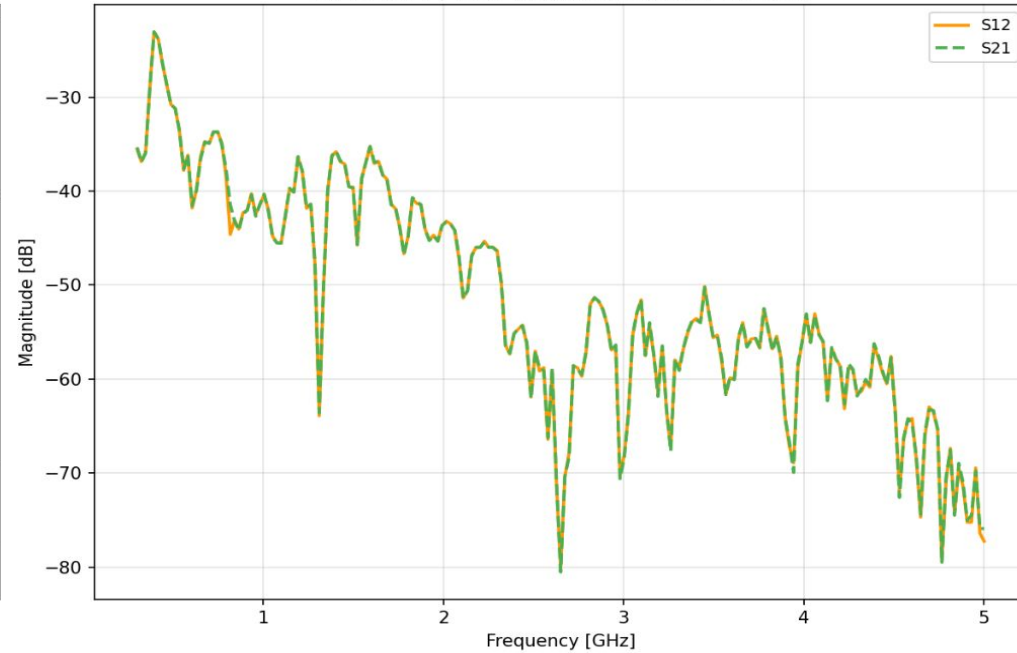
S12 & S21 | 5% graphite matching medium



S11 & S22 | 20% graphite matching medium

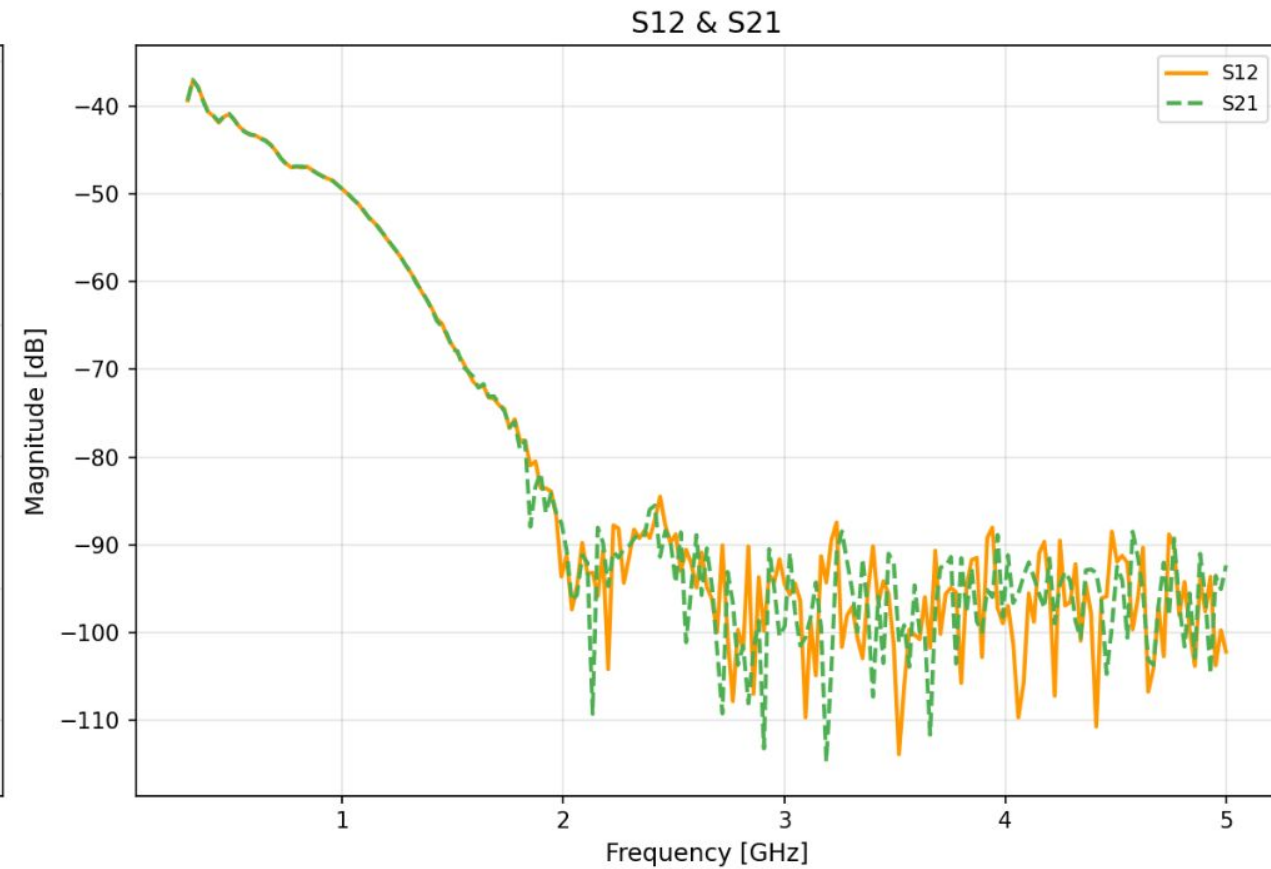
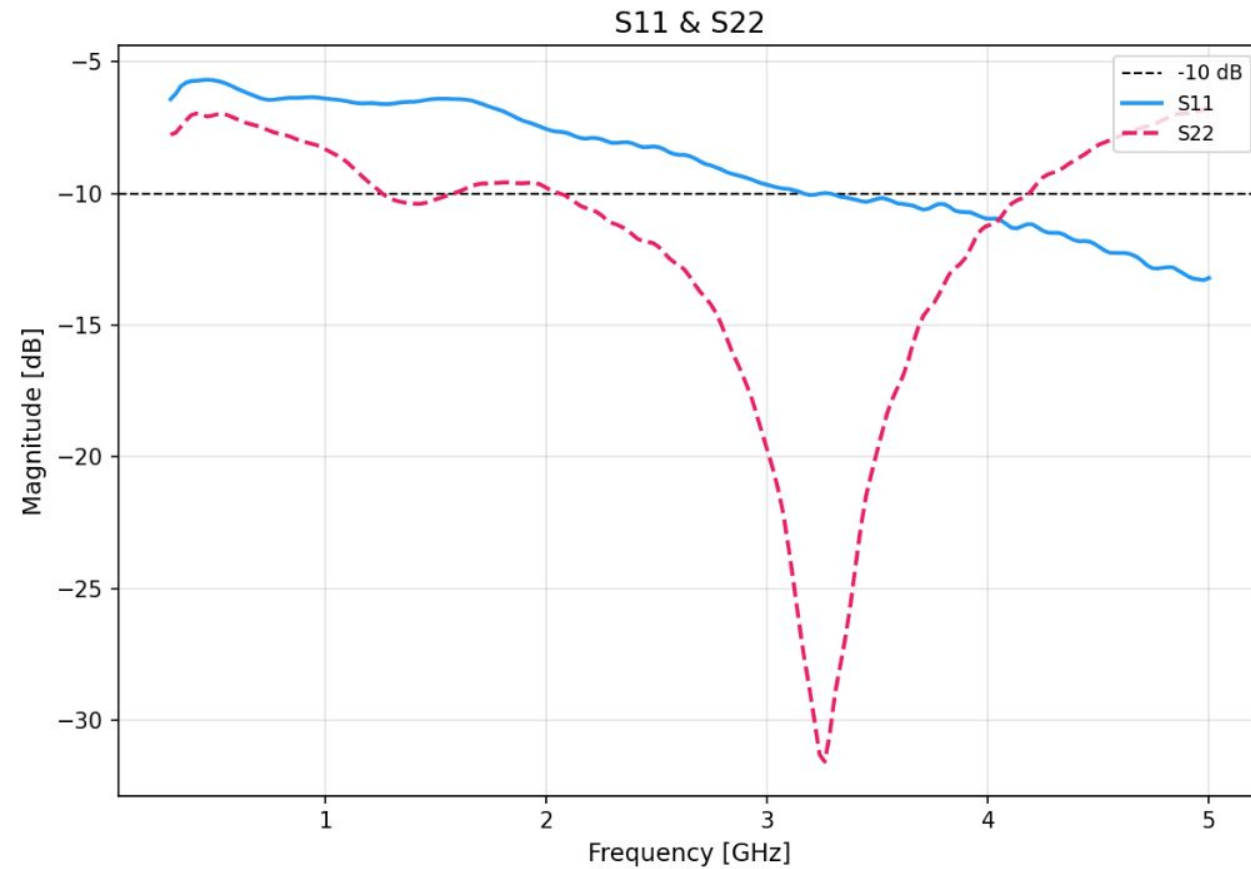


S12 & S21 | 20% graphite matching medium



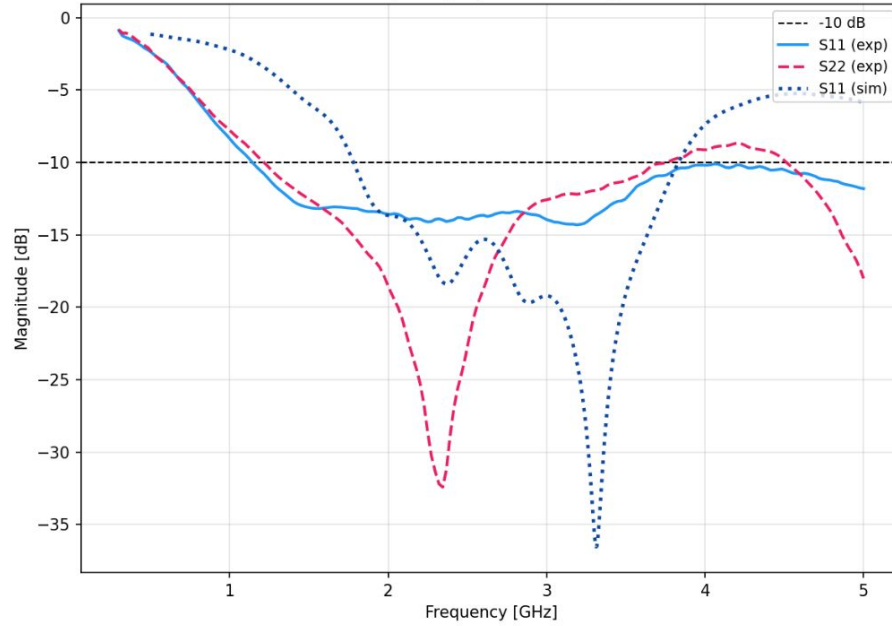
Source : Reflection and transmission parameters for the experiment on the Head Phantom with a coupling medium layer made of either 5 or 20 % graphite

myNeck — no matching medium

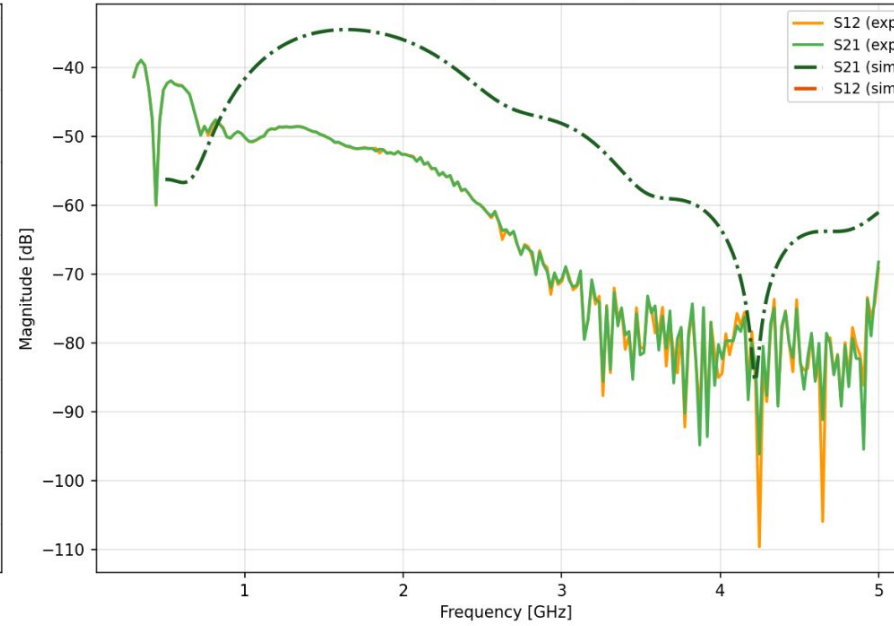


human neck — Graphite matching media comparison

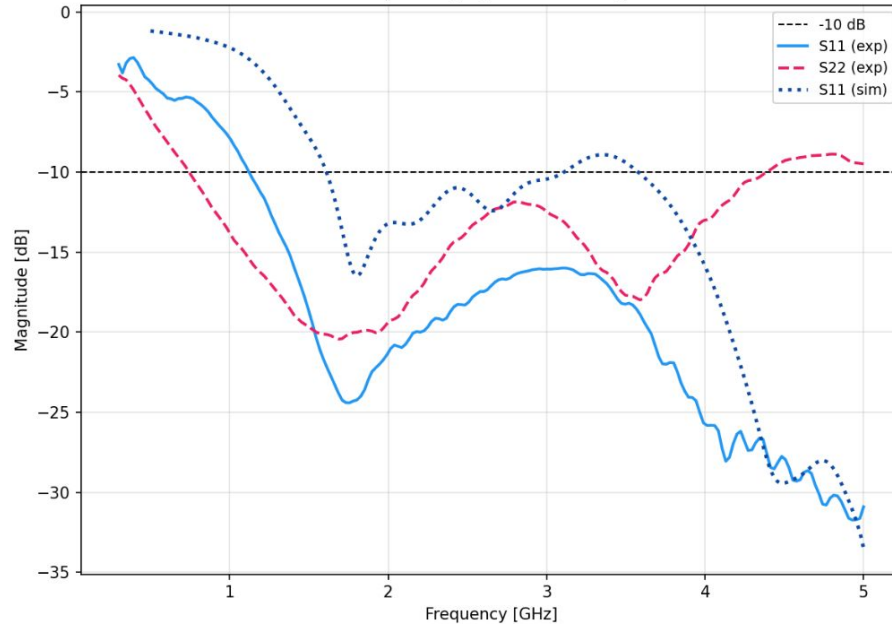
S11 & S22 | 5% graphite matching medium



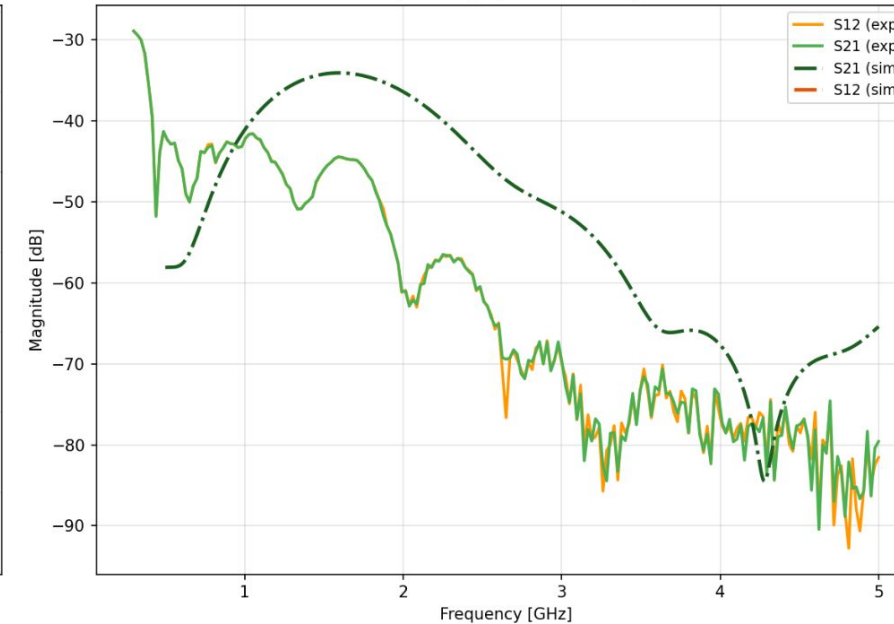
S12 & S21 | 5% graphite matching medium



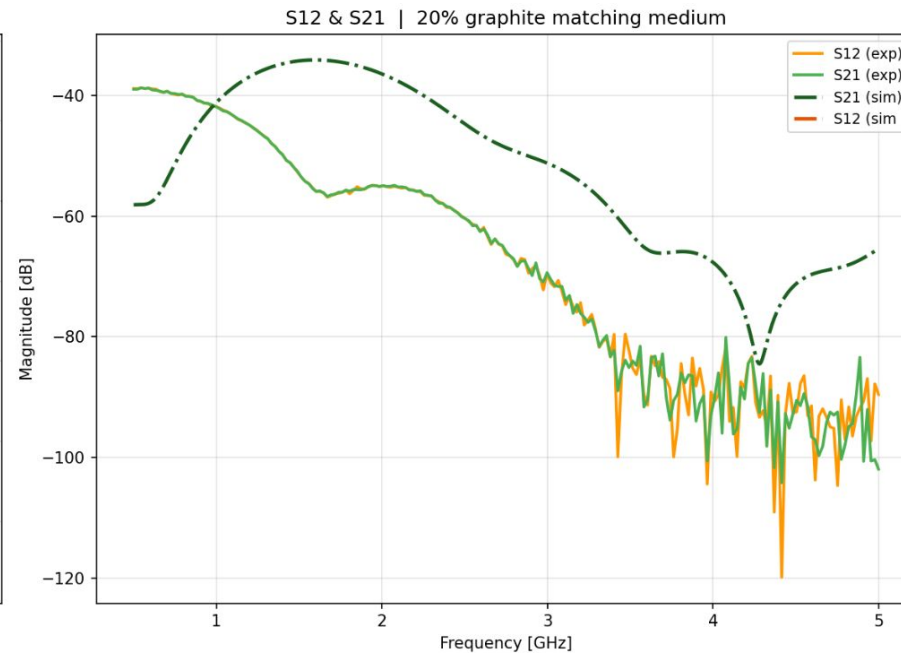
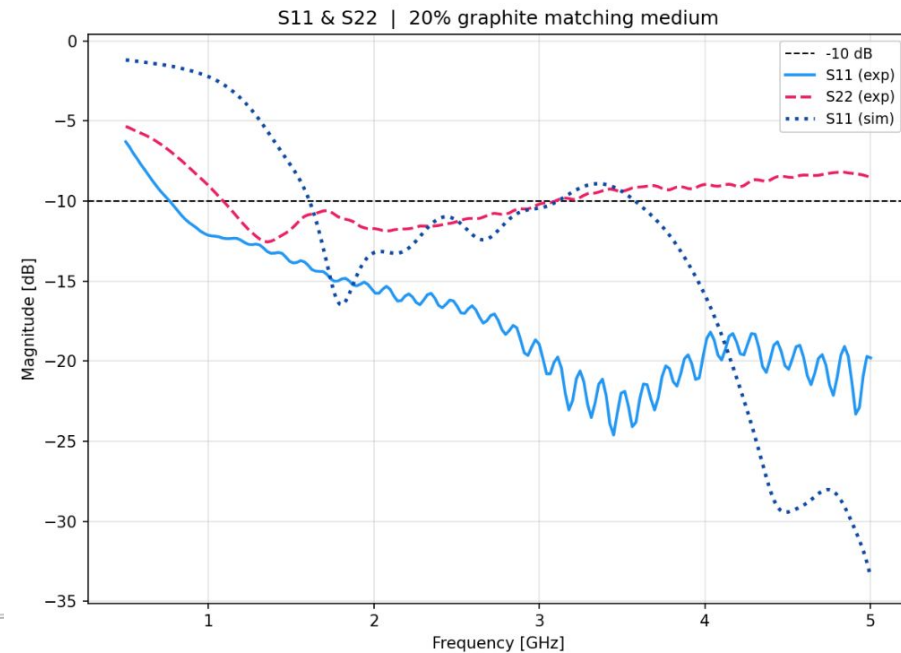
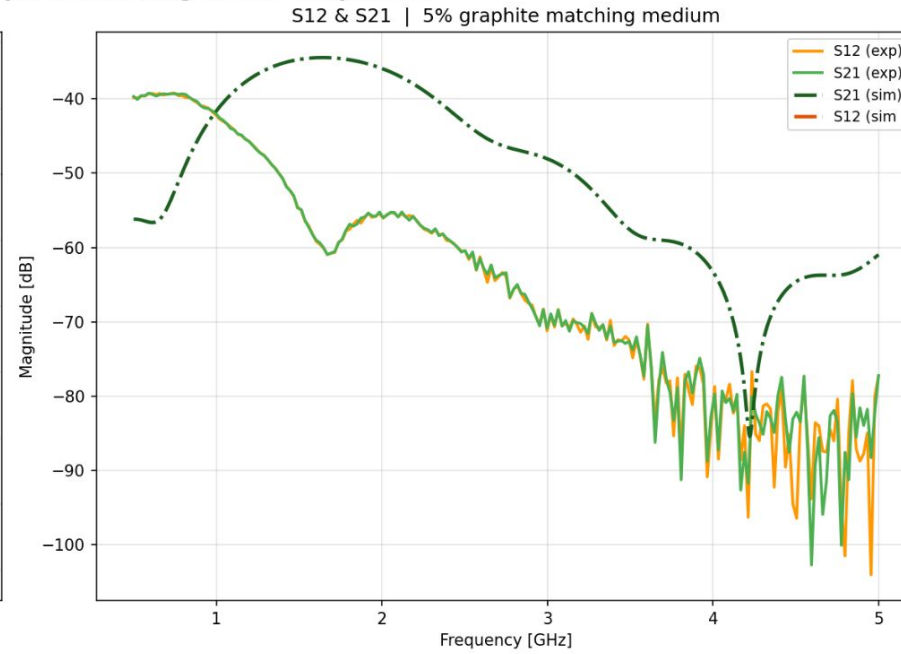
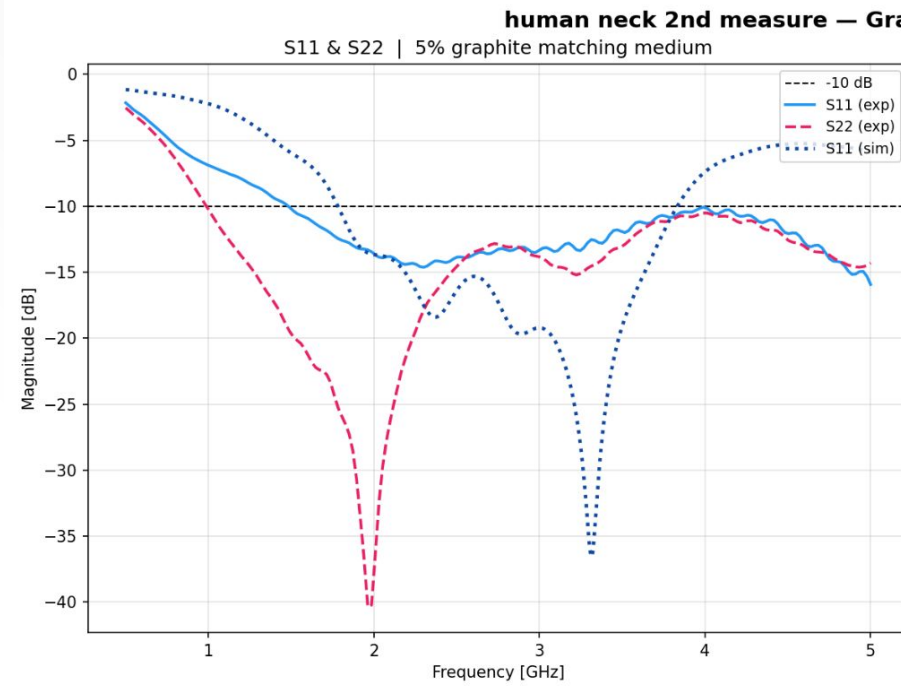
S11 & S22 | 20% graphite matching medium



S12 & S21 | 20% graphite matching medium



Source : Reflection and transmission parameters for the experiment on the human head with a coupling medium layer made of either 5 or 20 % graphite



Source : Reflection and
 transmission parameters for
 the experiment on the
 human head with a coupling
 medium layer made of either
 5 or 20 % graphite, 2nd
 batch of measurements

Citations

Slide 19:

- (1) Joshipura K, Muñoz-Torres F, Vergara J, Palacios C, Pérez CM. Neck Circumference May Be a Better Alternative to Standard Anthropometric Measures. *J Diabetes Res*. 2016;2016:6058916. doi: 10.1155/2016/6058916. Epub 2016 Feb 11. PMID: 26981543; PMCID: PMC4766356.
- (2) Hingorjo MR, Qureshi MA, Mehdi A. Neck circumference as a useful marker of obesity: a comparison with body mass index and waist circumference. *JPMMA. The Journal of the Pakistan Medical Association*. 2012 Jan;62(1):36-40. PMID: 22352099.

Slide 9:

- (2) K. Chang and L.-H. Hsieh, *Microwave Ring Circuits and Related Structures*, 2nd ed. Hoboken, NJ, USA: John Wiley & Sons, 2004.

Slide 27:

- [2] GeePs, "Simulation," 2023. [Online]. Available: <https://applis.iut.univ-paris-diderot.fr/CNRS/simulation.php>. [Accessed: Jul. 23, 2026].